IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000.

Digital Object Identifier 10.1109/ACCESS.2017.DOI

Benchmarking Deep Reinforcement Learning for Cartesian Path Tracking of Free-Floating Space Robot Manipulators

PATRICK S. ERMISCH[1], and ERIC GUIFFO KAIGOM[1]
[1]Department of Computer Science & Engineering, Frankfurt University of Applied Sciences, Frankfurt am Main, Germany
(e-mail: steven.ermisch@stud.fra-uas.de, kaigom@fra-uas.de)

Corresponding author: Eric Guiffo Kaigom (e-mail: kaigom@fra-uas.de).

ABSTRACT Free-floating space robots exhibit a strong dynamic coupling between ~~the~~ manipulator motion and base attitude disturbance, ~~complicating the~~ which complicates Cartesian path tracking of the end-effector. Supervised data-driven ~~approaches are challenging due to limited labeled data availability in space operations~~ motion control approaches are difficult to apply because labeled data from on-orbit operations are scarce. This paper therefore ~~explores trial-and-error learning techniques~~ investigates reinforcement learning for Cartesian path tracking of free-floating space robots in a physics-based Simulink/Simscape environment with integrated safety monitoring. ~~Since a~~ Because a unified quantitative and qualitative comparison of these methods is currently ~~missing in the sector of~~ lacking for on-orbit servicing, ~~we benchmark six Reinforcement Learning agents:~~ six RL algorithms are benchmarked, namely TRPO, TD3, SAC, PPO, ~~Policy Gradient~~PG, and DDPG. ~~We evaluate their performance using unified key performance indexes. These include~~ Their performance is evaluated using a unified set of key performance indicators covering training duration and stability, tracking accuracy, base attitude disturbance, motion smoothness, energy consumption, and collision ~~risks. We observe that PPO achieves the best overall~~ risk. Among the six algorithms, PPO yields the best trade-off between tracking accuracy, base disturbance minimization, and training stability. ~~Building on this insight, we quantify the~~ The effect of a systematic tuning of the neural network architecture and the key hyperparameters of PPO is then quantified for tracking both circular and ~~ramp-shaped~~ piecewise-linear Cartesian paths. Compared ~~to~~ with the default PPO configuration, the optimized version ~~significantly~~ improves the mean episodic return ~~by 80.7%~~from -2.23 to -0.43, reduces the mean squared tracking error by 67.7% and the maximum tracking error by 44.1%, and ~~stabilizes~~ reduces the base orientation ~~(~~error by 67.0%~~reduction). Moreover,~~ . In the 30 evaluation episodes considered, no collisions or joint-limit violations are observed for the optimized PPO ~~eliminates collisions and joint-limit violations while reducing~~ agent, and the energy consumption is reduced by 15.98%.

INDEX TERMS Free-floating on-orbit servicing, Proximal Policy Optimization (PPO), reinforcement learning, safe simulation, space robotics, trajectory tracking

I. INTRODUCTION AND BACKGROUND

Free-floating robot manipulators operating in ~~the space environment to inspect~~space for satellite inspection, capture, or ~~de-orbit satellites are expected to introduce unique control challenges when compared to~~ de-orbiting introduce control challenges not encountered with terrestrial robots [1]. In micro-gravity, the robot~~'~~'s base (typically a spacecraft) is not fixed and ~~thrusters are turned off~~the thrusters are inactive. Manipulator motions ~~can give rise to a change of~~ therefore change the position and orientation of the base, which in turn modifies the ~~final~~ pose of the end-effector. This reaction is ~~due to~~ a consequence of the conservation of ~~the~~ linear and angular momentum [2]. ~~The~~ Operating the manipulator under free-floating dynamics ~~of space robots has many benefits. An expensive consumption of fuel, such as hydrazine (N_2H_4), and an excitation of the robot dynamics for some activation frequencies of the pulse width modulation related to thruster control can be avoided~~ avoids the consumption of propellant (e.g. hydrazine, N_2H_4) and the excitation of structural modes by the pulse-width modulation of thruster control at certain

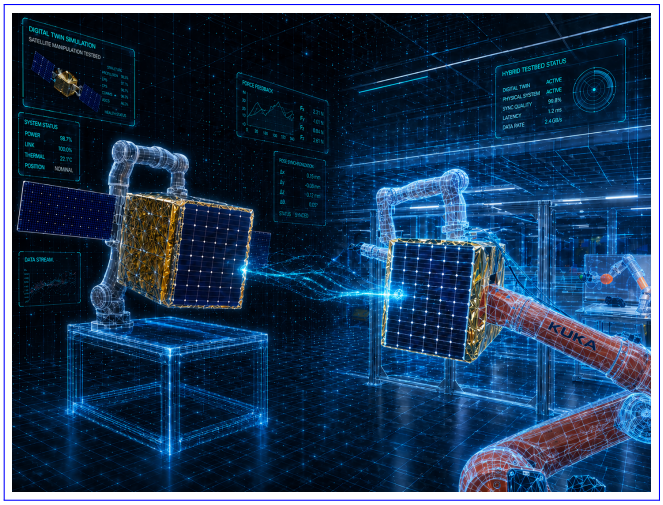


FIGURE 1: Virtual testbeds based on AI-driven digital twins support the safe and repeatable development of learning-based control methods for in-space operations under realistic yet controllable conditions. Illustration created using a picture of the physical setup and with the help of GPT-Image-2.

frequencies [3]. This ~~has the potential to extend the lifetime of space robots and enhance the~~ extends the operational lifetime of the spacecraft and supports mission success.

However, a free-floating base complicates ~~the achievement of robotized manipulations. Indeed,~~ manipulator control. Standard motion-planning approaches developed for fixed-base robots can fail to drive the end-effector ~~of the arm may fail to reach~~ to the desired target~~point if standard motion planning approaches dedicated to robots with a fixed base are simply applied. Further~~ . Additional constraints, such as communication latency and limited onboard computational resources, ~~make more intricate the~~ further complicate continuous real-time control and ~~stabilization of the base [1], [4]. In fact, traditional control~~ base stabilization [1], [4] . Model-based approaches that assume an accurate ~~model of the robot dynamics (e.g.~~dynamic model (e.g. reactionless planning or precise attitude control) ~~can struggle to adapt to uncertainties~~ are sensitive to model uncertainty and unmodeled dynamics. Supervised data-driven approaches ~~, on the other hand, are~~ are also difficult to apply ~~. Usually, the required~~ because labeled training data of sufficient amount and quality ~~of data are rather~~ are typically unavailable in the space sector. In ~~many cases, robotized tracking capabilities~~ addition, tracking capability under a free-floating ~~behavior of the arm's base need~~ base often has to be demonstrated ~~long~~ before the physical ~~robot~~ system is assembled and deployed [5]–[7].

These challenges motivate the development and simulation of ~~AI-empowered digital twins [8] under desired operational conditions to explore~~ AI-enabled digital twins (DTs) [8], as illustrated in fig. 1. Unlike standard DTs that primarily mirror sensed real-time data from physical counterparts, the DTs considered here combine physics-based simulation with learning components to contextualize system states and evaluate representative operating conditions. These conditions include uncertain dynamics, disturbances, faults, and contact-rich interactions associated with satellite capture. Faster-than-real-time simulation and systematic ablation studies can support controller design and help assess sim-to-real effects. In this role, AI-enabled DTs provide a training and validation environment for learning-based ~~control methods that can autonomously adapt to complex and unknown dynamicseven in the case of data scarcity~~controllers that combine safe exploration, policy optimization, and domain randomization under limited flight data and safety-critical constraints.

Reinforcement ~~Learning~~ learning (RL) has ~~emerged as a promising approach for~~ been increasingly applied in robotics, enabling agents to ~~learn~~ synthesize control policies through trial-and-error ~~interactions with entities, environments, and processes~~interaction with their environment. Recent advances in deep RL have ~~shown success in complex control tasks, but applying RL~~ addressed a range of complex continuous-control tasks, although applications to space robots ~~is still relatively novel~~ remain comparatively recent [9], [12]–[14], [16].

~~The work presented in this paper investigates the use of~~ This paper investigates continuous-action deep RL for ~~the tracking of a Cartesian motion of~~ Cartesian path tracking of the end-effector of a free-floating space manipulator. ~~It differs from previous contributions in three ways~~The contribution differs from prior work in three respects. First, ~~it focuses on a systematic comparison and explores the respective performance of~~ six different RL algorithms ~~beyond those mentioned e.g. in Table~~ 1~~, thereby quantitatively and qualitatively contributing to filling in a current gap in space robotics~~are systematically compared (see section VI) and complemented by ablation studies (see section VII-B), addressing a gap left by the works listed in Table 1. Second, ~~several constraints ,~~ constraints such as collision avoidance ~~,~~ are monitored through formal methods ~~applied from a software-enabled and practical perspective. Finally, this work investigates the minimization of the base attitude~~ integrated into the simulation toolchain (see section V). Third, base attitude minimization during arm motion is treated as an explicit control objective in conjunction with ~~constraint monitoring previously mentioned. Recall that the~~ the constraint monitoring above. The induced base disturbance is ~~omitted in most recent contributions, as highlighted~~ not addressed in most of the works listed in Table 1. ~~We are not aware of any comprehensive work that compares~~ To the authors' knowledge, a unified comparison of RL algorithms for ~~the~~ Cartesian path tracking of free-floating space robots ~~while considering the~~ that jointly considers base attitude minimization and ~~unifying constraint monitoring by leveraging formal methods, to the best of our knowledge (see, also)~~formal-methods-based constraint monitoring remains unavailable.

TABLE 1: Recent advances in Deep RL-based motion management for free-floating space robots. Abbreviations: RL Alg. = reinforcement learning algorithm; DOF = degrees of freedom; Obs./Act. Dim. = observation-/action-space dimension; vel. = joint velocities; rates = joint rate commands; pos./att. = position/attitude; pos./ori. = position/orientation; EE = end-effector; Formal Ver. = formal verification; Base Dist. Min. = base disturbance minimization; d_{safe} = minimum admissible distance; $\tau_{\max}$ = torque bound; K1–K9 = evaluation KPIs (see Section IV); N/R = not reported.

Contribution	RL Alg.	Task Objective	DOF	Simulator	Obs./Act. Dim.	Safety Limits	Eval. Metrics	Formal Ver./ Base Dist. Min.
[9], 2023	SAC	Obstacle avoidance	7	OpenAI Gym	14 / 7D vel.	Reward-based	Success rate	No/No
[10], 2024	DDPG	Collision avoidance	6	Custom	N/R / 6D vel.	Reward-based	Success rate	No/No
[11], 2024	PPO	Joint failure adaptation	7	MATLAB Simscape + SPART	32 / 7D rates	Pos./att. threshold	Per-joint success rate	No/No
[12], 2024	DDPG	Path planning	6	Custom	32 / 6D vel.	Reward-based	Convergence, task completion	No/No
[13], 2025	PPO	Trajectory planning	6	PyBullet	N/R / 6D	Joint limits	Success rate, pos./ori. error	No/No
[14], 2025	LSAC	Trajectory post-processing	N/R	Custom + air-bearing	N/R / N/R	Uncertainty bounds	Tracking error	No/No
[15], 2025	DDPG	Trajectory planning	N/R	Custom	N/R / N/R	Base dist. (soft)	Reward, base dist. red.	No/Yes
Ours	**PPO**	**Cartesian EE tracking**	**4**	**MATLAB/Simulink + Simscape**	**23 / 4D τ**	d_{safe}, $\tau_{\max}$, joint limits	K1–K9	**Yes/Yes**

As shown in Table 1, the surveyed prior contributions typically evaluate a single RL algorithm on a specific subtask, which limits conclusions about the relative merits of different learning paradigms for free-floating space robot control. Furthermore, the surveyed literature does not jointly address three compounding challenges: (i) a multi-algorithm benchmark under strictly unified conditions, (ii) explicit minimization of base attitude disturbance as a co-equal control objective alongside end-effector tracking, and (iii) integration of formal safety verification directly into the RL training loop. The present work addresses these gaps jointly by providing a reproducible and safety-aware comparison framework that extends the individual studies listed in Table 1.

The ~~key contributions and novelties~~ main contributions of this work ~~include~~are summarized as follows:

- ~~A comparative evaluation of RL Agents: We implement and evaluate six state-of-the-art continuous RL algorithms ,~~ *Comparative evaluation of RL agents.* Six continuous-action RL algorithms are implemented and evaluated on the same path-tracking task. The algorithms are Trust Region Policy Optimization (TRPO) [17], Twin Delayed Deep Deterministic Policy Gradient (TD3) [18], Soft Actor-Critic (SAC) [19], Proximal Policy Optimization (PPO) [20], vanilla Policy Gradient (PG) [21], and Deep Deterministic Policy Gradient (DDPG) [22]~~, on the same path tracking task of a free-floating robot. This side-by-side comparison~~ . The comparison is performed under unified conditions ~~reveals which algorithm is likely to deliver the best possible performance when it comes to motion tracking of a free-floating space robot and why~~and identifies which algorithm offers the most favorable trade-off between tracking accuracy, base disturbance, and training stability.
- ~~An integration of formal safety measures: To meet safety-critical space operation requirements, we incorporate formal methods~~ *Integration of formal safety mechanisms.* Formal methods are integrated into the learning loop [2], [23]–[25]. A collision ~~monitoring, momentum conservation check , and formal verification of joint torque limits are embedded in the simulation. This integration ensures the RL agent cannot violate fundamental and critical~~ monitor and a momentum-conservation check run at runtime, and the joint-torque bound is additionally verified offline with a model checker on a bounded abstraction of the policy. This combination monitors the targeted safety properties (~~like avoiding collisions or exceeding actuator limits~~collision avoidance and bounded actuator commands) during training and testing.
- ~~A high-fidelity simulation environment: We develop a high-fidelity~~ *Simulation environment for multi-criteria evaluation in on-orbit servicing.* A MATLAB/Simulink simulation ~~of~~ framework is developed for a free-floating space robot ~~, using a URDF-based~~ based on a URDF-derived multibody model and the Simscape physics engine. The robot ~~, a cube-shaped base with a 4-Degree of Freedom (DOF) robotic arm, is modeled with realistic kinematics, dynamics~~consists of a cuboid base and a 4-degree-of-freedom (4-DOF) revolute arm, modeled with consistent kinematic, inertial, and sensor/actuator parameters~~and features. A dedicated reinforcement learning environment provides the agent with~~ . The associated RL environment provides observations and rewards ~~, including a specially designed~~ through a composite reward function that ~~encourages accurate~~ penalizes end-effector ~~trajectory tracking while penalizing base movement~~tracking error, base motion, large joint velocities, and excessive control effort~~(energy). Through reward shaping and carefully chosen~~ . Together with the reset conditions, ~~the agent~~

learns to accomplish the task without causing excessive base disturbance or safety violations this reward shaping allows the agent to learn the task while keeping base disturbance and safety violations bounded [22], [26]–[29].

- The optimization of the PPO agent : *Optimization of the PPO agent.* Based on the comparative results, we identify PPO as the top- performing algorithm and further refine it for path tracking tasks. We adjust the PPO is selected for further tuning. The neural network architecture and tune hyperparameters (e.g., key hyperparameters (learning rate, mini-batch size, clipping ratio, entropy regularization, etc.) and entropy regularization) are adjusted to improve training stability and policy performance. The optimized PPO agent achieves significantly better tracking accuracy, reduces tracking error, produces smoother control inputs, and zero safety violations , outperforming the shows no safety violations in the reported evaluation episodes relative to the default configuration on all key metricsthe tested KPIs.

Overall, our study shows that a combination of multiple-body

The integration of multibody simulation, deep RLreinforcement learning, and formal verification can yield a robust and safe control policy for a free-floating space robot. In the following, we first describe the system modeling and RL framework, then detail the formal safety components, present a comparative analysis of the RL agents, and finally discuss the optimized PPO agent 's performance, before concluding with broader insights and future directionsconsidered here yields control policies that respect the imposed safety constraints while reducing tracking error and base disturbance relative to the default configuration. The remainder of the paper is organized as follows. Section III describes the system model and the simulation environment. Section IV introduces the reinforcement learning framework. Section V presents the safety monitors and formal-methods components. Section VI reports the comparative evaluation of the six RL algorithms. The optimization of the PPO agent and its evaluation on circular and piecewise-linear references follow, before discussion and conclusion.

II. RELATED WORK

A. REINFORCEMENT LEARNING FOR ROBOTIC CLASSICAL CONTROL TECHNIQUES FOR SPACE ROBOTS

Model-based approaches have historically dominated space robot control. Computed torque control and resolved-motion rate methods leverage accurate dynamic models to achieve precise end-effector positioning [1]. Reactionless path planning explicitly minimizes base disturbance by exploiting the robot's redundancy and a dynamics-related null-space to design joint velocities, while attitude regulation methods use reaction wheels or thrusters to counteract momentum transfer [2], [7]. These approaches offer strong performance guarantees and interpretability when the system model is accurate. However, they are sensitive to model uncertainties and unmodeled dynamics, require significant engineering effort for model identification, and are ill-suited to the data-scarce, hazardous conditions of on-orbit operations.

B. REINFORCEMENT LEARNING-BASED CONTROL TECHNIQUES

RL has demonstrated success in various robotic manipulation tasks [22], [30]. Policy gradient methods (PG, PPO, TRPO) and actor-critic algorithms (DDPG, TD3, SAC) have been widely applied to continuous control problems [17]–[21].

C. CONTROL OF FREE-FLOATING SPACE ROBOTS

Traditional approaches for space robot control include computed torque control, reactionless path planning, and attitude regulation [1], [2]. Recent works have started exploring RL Applied to free-floating space robots, RL-based methods have been investigated for path planning [12], [16], obstacle avoidance [9], and trajectory tracking [13]. , and failure management [11]. RL does not require an explicit dynamic model of the system, which is relevant when model parameters are uncertain or unavailable. However, current RL-based approaches share recurring limitations. Most studies evaluate only a single algorithm, address a single objective, omit base attitude minimization, and do not provide systematic benchmarking under comparable conditions.

C. SAFE REINFORCEMENT LEARNINGHYBRID SAFETY-AWARE CONTROL SYSTEMS

Safe RL integrates safety constraints through formal verification, barrier functions, and constrained optimization [24], [25], [31], [32]. However, application to space robotics with formal safety guarantees remains limitedShielding approaches [25] intercept unsafe actions at runtime and substitute corrective commands, while control barrier functions (CBFs) [32] encode safety as a continuously enforced constraint on the policy's action set. These hybrid methods combine the adaptability of RL with formally specified runtime constraints, helping to enforce critical properties such as joint limit compliance and collision avoidance. However, their application to free-floating space robots remains largely unexplored: existing works rarely integrate formal verification directly into the RL training loop or combine it with base attitude minimization objectives.

D. RESEARCH GAPNEED FOR BENCHMARK STUDIES

Despite recent advances, systematic comparisons Comparative evaluations of RL algorithms for free-floating space robot control are lacking. Most prior works focus on single algorithms and omit base attitude minimizationor scarce. Table 1 surveys recent deep RL contributions to free-floating space robot motion management. As shown, all surveyed works address only individual goals with a single algorithm. None simultaneously combines

multi-algorithm and multi-criteria benchmarking, base disturbance minimization, and formal safety verification. This absence of unified evaluation makes it difficult to draw conclusions about relative algorithm performance and to guide algorithm selection for future missions.

E. RESEARCH GAP

The preceding survey reveals three mutually reinforcing gaps in the existing literature. First, no prior work systematically compares multiple RL algorithms under identical conditions for Cartesian end-effector tracking of free-floating space robots. Second, base attitude minimization, a safety-critical requirement in on-orbit operations, is omitted in most RL-based contributions (see Table 1). Third, the integration of formal safety verification directly into the RL training and evaluation loop has not been demonstrated for this class of systems. The present work addresses all three gaps simultaneously. It benchmarks six representative RL algorithms under unified conditions, incorporates base attitude minimization as an explicit control objective, and integrates formal safety monitoring into the RL training and evaluation loop, with the actuator-torque bound additionally established by offline model checking.

III. SYSTEM MODELING AND SIMULATION ENVIRONMENT

A. ADVANTAGES OF AI-ENDOWED DIGITAL TWIN TECHNOLOGY IN ON-ORBIT SERVICING

Recent advances in AI-driven ~~digital twinning~~ DTs have demonstrated considerable benefits for on-orbit servicing, ranging from automated lifecycle management of satellites to the orchestration of skillful agents that autonomously carry out complex assembly, integration, and testing tasks [33], [34]. By combining contextualized real-time sensor data with AI ~~that accommodates uncertainties, digital twins~~ methods that account for uncertainty, DTs bridge the physical and digital ~~world with accelerated~~ domains and enable simulation-driven analysis under ~~desired operational conditions and at low costs~~diverse operational conditions at reduced cost. Verification and validation also benefit from this paradigm, as changes in system design or operational scenarios can be described in scenario spaces and integrated with ~~digital twins~~ DTs for flexible verification [35].

In this work, the information model of the DTs associated with the target satellite and the chaser space robot provides a structured and interoperable description of the satellite and robotic systems, including their kinematics, inertial properties, joint configurations, and physical constraints. The AI capability is embodied by a deep reinforcement learning agent, specifically PPO, which is trained and evaluated through interaction with the corresponding DTs and additional DTs involved in representative in-space operation scenarios. This distinguishes the proposed approach from classical simulation. Rather than relying on pre-specified controllers, the robotics-related DT hosts a learning-based agent that autonomously synthesizes control strategies from experience. In this way, the DTs are not limited to reproducing predefined system behavior, but serve as adaptive training and validation environments that support the evolution of control strategies under varying operational conditions. During training, scenarios and constraints are sampled from bounded uniform distributions, allowing the agent to explore a broad range of admissible operating conditions. This improves robustness to modeling errors, disturbances, and partially unknown dynamics without requiring explicit model knowledge.

B. ROBOT MODEL

The virtual testbed ~~for this research allows for the simulation of~~ simulates a free-floating space manipulator ~~inspired by~~ derived from the Spacecraft Robotics Toolkit (SPART) example [36]. The robot consists of a cuboid base (serving as the spacecraft) ~~with~~ and a 4-degree-of-freedom (4-DOF) serial manipulator ~~arm mounted on it~~ (see Figure 2). The arm has four revolute joints and four links~~, and its~~. Its kinematic structure and inertial properties are ~~defined~~ specified in a Unified Robot Description Format (URDF) file (~~SpaceRobot.urdf) which is~~ `SpaceRobot.urdf`) imported into Simulink~~'s multibody simulation~~'s multibody environment [26]–[28]. ~~Identified values~~ The geometry and the inertial properties of the links are taken from identified values reported in the literature (see, ~~.~~e.g., [37]) ~~were chosen for the geometry and the inertial properties of links of the space arm considered in this work and~~ and stored in the URDF model. The ~~result is a realistic multibody model with a floating base and an arm, capturing the strong~~ resulting multibody model captures the dynamic coupling between ~~them~~ the floating base and the arm [2].

C. DYNAMIC SIMULATION IN MATLAB/SIMULINK

The simulation is implemented in Simulink using the Simscape Multibody physics engine. Gravity is ~~turned off (~~set to $[0\ 0\ 0]^T$ ~~)~~ to emulate the microgravity orbital environment. The base is connected to the inertial world frame through an unactuated ~~6-DoF~~ 6-DOF joint [38], allowing ~~it to move freely translate and rotate~~free translation and rotation. No external forces or torques act on the system~~. Thus, the base only moves~~, so the base moves only in response to ~~forces~~ the internal torques generated by the manipulator ~~'s joints. This configurationensures that momentum is conserved, motions of the arm can lead to a reaction of the~~ joints. Under this configuration, the total linear and angular momentum are conserved, and arm motions induce a reactive motion of the base [2], [7], [29], [32], [38].

The simulation operates under a set of idealizing assumptions that are standard in RL-based space robotics research and are explicitly acknowledged here to facilitate reproducibility and fair comparison. First, ideal sensing is assumed. All state variables (joint angles, joint velocities, base orientation quaternion, and base angular velocity) are read directly from Simulink signal buses without sensor

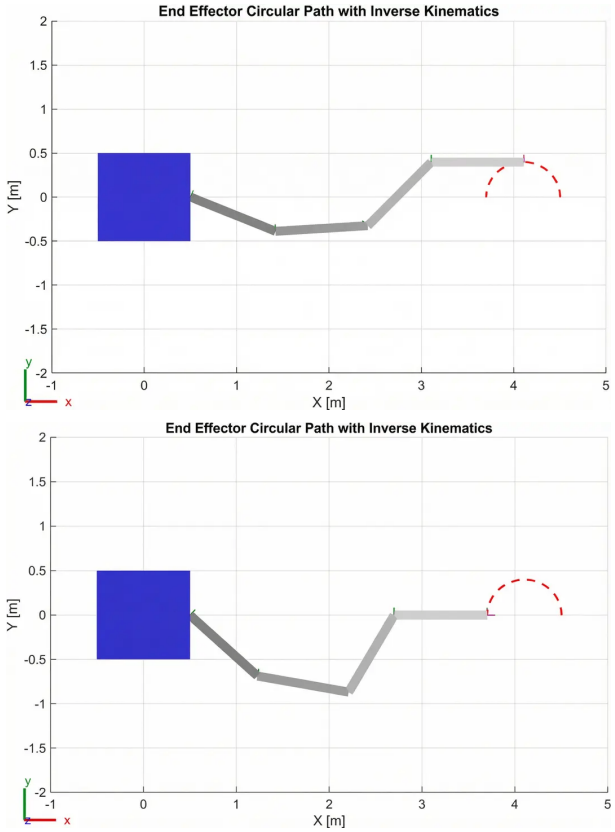


FIGURE 2: Tracking the desired end-effector circular path using inverse kinematics.

noise, quantization errors, or measurement bias. Second, zero communication latency is assumed. The RL agent receives observations and issues torque commands within the same discrete time step, with no transmission delay between sensing and actuation. Third, the robot is modeled as a rigid multibody system. Flexible link dynamics, joint compliance, and thermal drift are not included. Fourth, no external disturbances act on the spacecraft. Gravity gradients, atmospheric drag, solar radiation pressure, and residual magnetic torques are all set to zero, consistent with the free-floating assumption in low-Earth orbit. These assumptions establish a clean, controlled benchmark environment in which performance differences can be attributed unambiguously to algorithmic choices rather than to environmental variability. Their impact on real-world deployment is discussed in the Limitations section.

The choice of MATLAB/Simulink as the simulation backbone was made deliberately in view of the alternatives commonly used in robotics and reinforcement learning research, namely Gazebo [49] and MuJoCo [48]. MuJoCo offers a highly optimized contact and multibody solver and has become a de-facto standard for RL benchmarks in continuous control [50]. However, it lacks a native interface to Simulink, Stateflow, and the model-based formal verification ecosystem that this work relies on to express and analyze safety monitors. Gazebo provides mature ROS integration, plug-in-based sensor models, and is well suited for hardware-in-the-loop development [51], but its general-purpose multibody back-ends are typically less convenient for free-floating spacecraft scenarios and likewise offer no native path to formal verification artifacts. Simscape Multibody, in contrast, delivers a validated multibody solver for the free-floating base–manipulator coupling and integrates seamlessly with Stateflow and the Simulink Design Verifier used to specify the runtime safety monitors that constitute a central contribution of this work. Combined with the Reinforcement Learning Toolbox and a deterministic fixed-step solver, this yields a single, reproducible toolchain from physical model through learned policy to formally checked safety logic. Table 2 summarizes the trade-off. Faster raw simulation throughput in MuJoCo and richer sensor and ROS tooling in Gazebo are exchanged here for tight coupling between the dynamics, the RL agent, and the formal-methods pipeline.

The Simulink model is composed of several subsystems, as illustrated in Figure 3. The Robot Plant subsystem computes the ~~full 6-DOF~~ multibody dynamics of the free-floating robot at each time step (see Figure 4) ~~, taking joint torque inputs and outputting the resulting robot states~~ by numerically integrating the equations of motion for the applied joint torques [2]. The Controller subsystem executes the learned RL policy~~, applying joint torque commands that~~ ; the

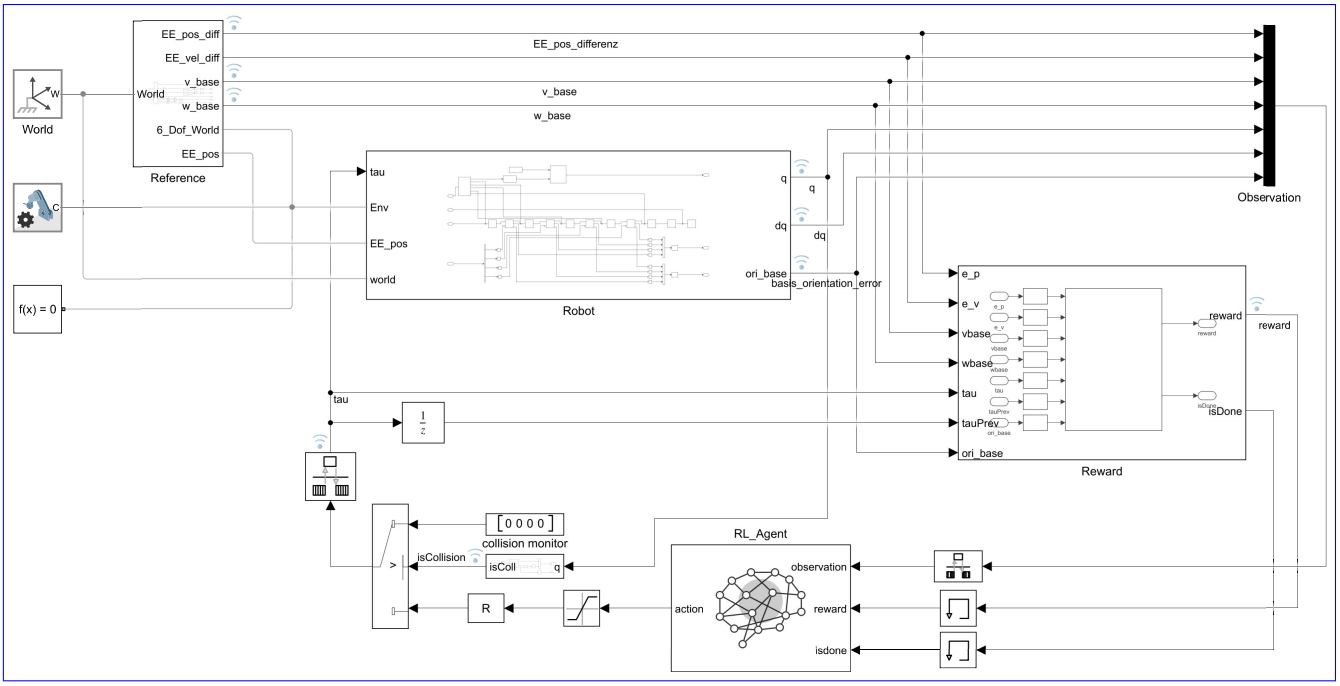


FIGURE 3: Overview of the Simulink model `SpaceRobot.slx`
This figure shows the Simulink-based closed-loop simulation framework for the free-floating space robot. The model integrates the robot dynamics, reference generation, observation assembly, reward computation, and a PPO-based reinforcement learning agent that outputs joint torques. The reward and termination logic evaluate end-effector tracking performance, base motion disturbance, and control effort to guide policy learning.

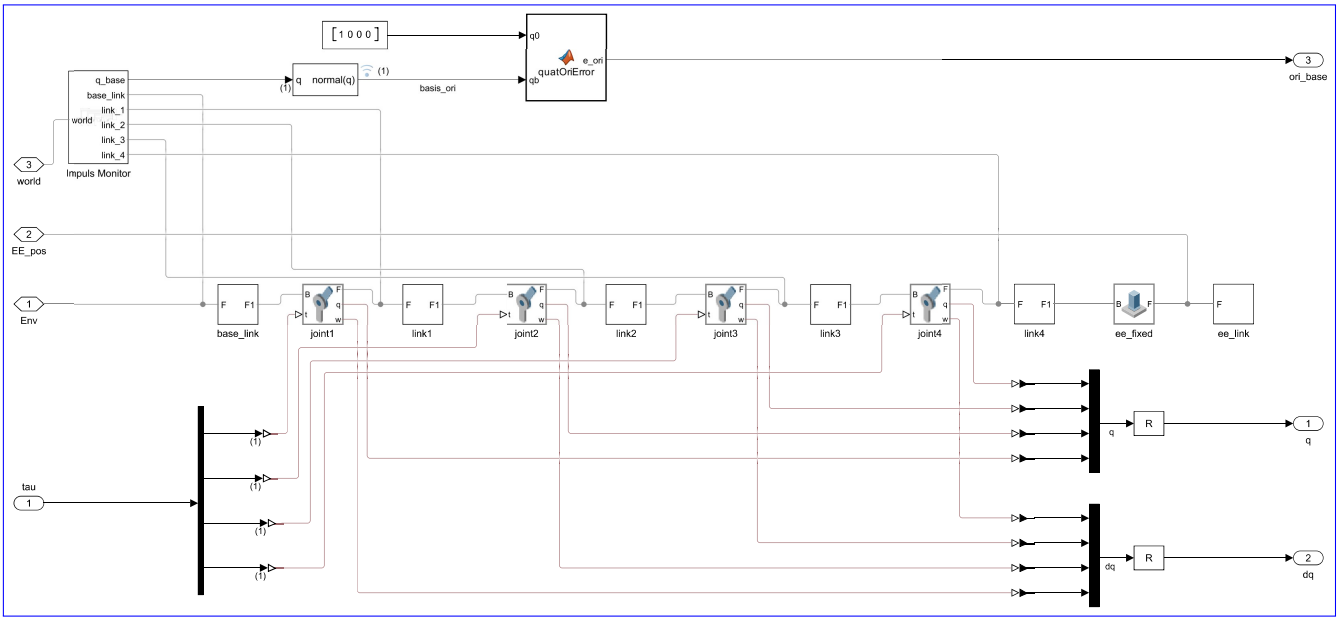


FIGURE 4: Overview of the Robot Subsystem in the Simulink Model `SpaceRobot.slx`
This figure depicts the Simulink/Simscape Multibody model of the free-floating space robot, consisting of a floating base and a serial 4-DOF manipulator. Joint torque commands are applied to the revolute joints, while joint positions and velocities are measured and fed back as state variables. The base orientation is represented as a normalized quaternion and compared to a reference to compute the base attitude error.

commanded joint torques pass through saturation and safety verification blocks before reaching the plant. ~~If~~ When the

TABLE 2: Comparison of simulation platforms relevant to free-floating space robotics with formal safety monitors.

Criterion	MATLAB/ Simulink	Gazebo	MuJoCo
Physics engine	Simscape Multibody	ODE, Bullet, DART	MuJoCo (native)
Free-floating multibody fidelity	High	Medium	High
Formal verification integration	Native (Stateflow, Design Verifier)	None	None
RL framework	RL Toolbox	gym-gazebo, ROS bridges	Gymnasium, dm_control
Sensor and ROS integration	Limited (via add-ons)	Extensive	Limited
Code generation / HIL path	Mature (Embedded Coder)	Via ROS 2	Limited
Licensing	Commercial	Open source	Open source

collision monitor signals reports an imminent collision, this block the controller issues an emergency stop and gradually overrides all overrides the torques to zero [24], [25]. The Reference Trajectory Generator provides the desired end-effector trajectory and base attitude (see Figure 5), outputting the current target position and velocity at each time step. Various A set of sensors measure the robot 's measures the robot state (joint angles, joint velocities, base orientation, and base angular velocity) , combining them into an and assembles the observation signal for the RL agent (detailed in section IV). Finally, safety monitors enforce constraints. For instance, a Collision Monitor checks distances between robot components the operational constraints: a collision monitor compares the distance between selected robot components against a threshold d_{safe} and triggers a safety halt if any the distance falls below a thresholdd_{safe}this threshold, while a joint-limit monitor triggers an emergency stop if any joint exceeds its allowed admissible range [24], [25], [39].

To accurately simulate couple the continuous dynamics while interfacing with the discrete-time RL agent, a consistent integration scheme is used. The Simscape solver is configured to operate with a fixed-time step with a fixed-step size equal to the agent''s decision interval (Δt) ,instead of the default variable-step , to ensure solver, ensuring synchronization between the physics simulation and the RL decision-makingcycle. Rate transition blocks handle any rate differences , avoiding aliasing and ensuring that state observations and action commands align appropriately the remaining rate differences and prevent aliasing between the continuous plant and the discrete agent loop [40], [41]. Additionally, Simulink 's Simulink unit consistency checks are enabled to catch any modeling errors (e.g. mismatched units) detect modeling errors such as mismatched units early in development.

D. REFERENCE TRAJECTORY AND DESIRED TASK

The target task for the robot task is to follow a periodic end-effector trajectory while minimizing the base disturbance. As previously mentioned, the The reference trajectory is chosen as a circular path (see Figure 2). Prior to trainingthe RL agents, the trajectory is was validated through inverse kinematics to ensure it is geometrically feasible confirm geometric feasibility for the manipulator (within compliance with joint limits and avoiding absence of singularities). The inverse kinematics solution produced inverse-kinematics solution yielded a consistent set of joint movements that achieve trajectories that realize the desired end-effector motion, confirming that the circle can be tracked by the arm without requiring unrealistic motions from a kinematic viewpointcircular path is kinematically feasible. This precaution guarantees ensures that the task is achievable , at least by some by at least one controller, so if that any failure of the RL agent fails to learn it, it is due to learning limitations rather than an unattainable desired Cartesian poses of the end-effectorcan be attributed to learning, not to a kinematically infeasible reference. During simulation, the reference subsystem outputs the desired end-effector position and velocity as time series $EE_{ref}(t)$ and $EE_{vref}(t)$ in the workspace $EE_{\text{ref}}(t)$ and $EE_{v,\text{ref}}(t)$ at each time step. From these terms, the instantaneous tracking error signals are computed for the The instantaneous position and velocity of the end-effector. These error signals are key components of the observation and are also used in tracking errors are derived from these signals. They enter both the observation vector and the reward computation that penalizes large errors.

IV. REINFORCEMENT LEARNING FRAMEWORK

State (Observation) Space: The **State (Observation) Space.** At each time step the RL agent receives an observation capturing the system's key state information relevant to control at each time step that summarizes the state variables relevant for control [22]. The state vector is composed consists of the following elementscomponents:

- End-Effector Tracking Error: The *End-effector tracking error.* The Cartesian position error $e_p \in \mathbb{R}^3$ between the current and reference end-effector 's current position and the reference target position (in Cartesian coordinates). If positions, together with the velocity error $e_v \in \mathbb{R}^3$. When the end-effector is exactly on the desired trajectorypoint, $e_p = 0$. Additionally, the error in end-effector velocity $e_v \in \mathbb{R}^3$ (difference between current and target end-effector velocities) is included. Together, ($e_p$on the reference trajectory, e_v) provides the agent a direct indication of tracking performance. Specifically, it tells the agent how far off the trajectory the end-effector is and whether it is lagging or overshooting in velocity $e_p = 0$. The pair (e_p, e_v) indicates both the spatial deviation from the reference and the velocity mismatch.
- Arm Joint States: The positions and velocities of each of the 4 arm joints. We include the *Arm joint states.* The

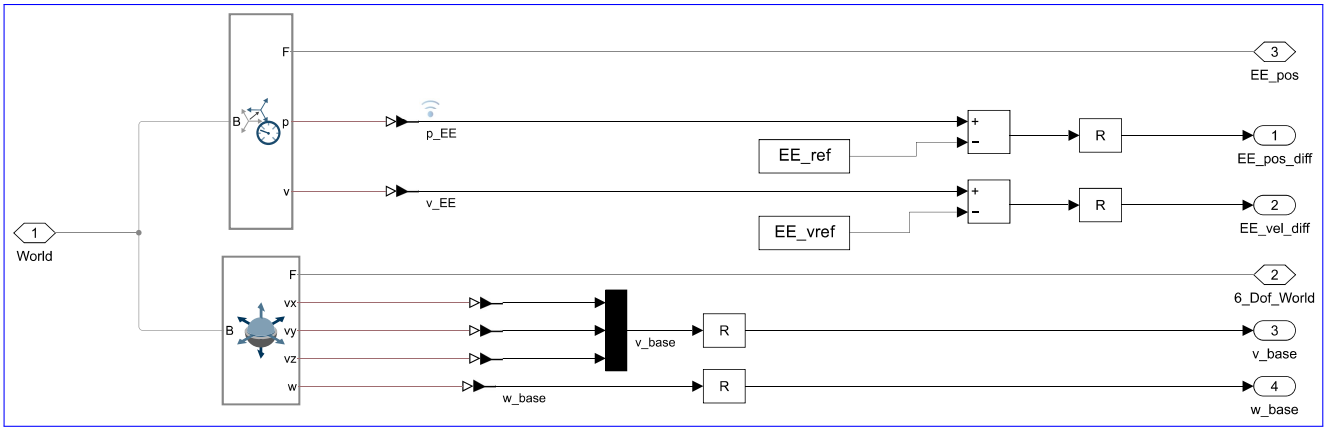


FIGURE 5: Overview of the Reference Subsystem in the Simulink Model `SpaceRobot.slx`
This figure shows the computation of end-effector position and velocity tracking errors with respect to the reference trajectory. The measured end-effector states are compared to the reference signals to generate position and velocity error vectors used for observation and reward calculation. In addition, the base linear and angular velocities are extracted to capture base motion induced by manipulator actuation.

joint angles $q \in \mathbb{R}^4$ and joint angular velocities ~~$\dot{q} \in \mathbb{R}^4$ in the observation. These inform the agent about the manipulator's~~ $\dot{q} \in \mathbb{R}^4$. These describe the configuration and motion state ~~(e.g., at rest or not). (Note: Joint angles can be normalized or wrapped as needed, in our implementation they~~ of the manipulator. Joint angles are provided in radians ~~along with known joint limits.)~~ together with the corresponding mechanical limits.

- ~~Base Motion: The free-floating~~ *Base motion*. The orientation, angular velocity, and linear velocity of the free-floating base. The base ~~'s orientationand angular velocity. We parameterize the base's orientation using quaternions. The~~ attitude error is computed from the measured base quaternion q_b and the reference quaternion ~~$q_{base} \in \mathbb{R}^4$ describes the orientation of the base , capturing its attitude error from a reference orientation.~~ q_0 as the relative quaternion $q_{\text{err}} = q_0^* \otimes q_b$ (with sign convention enforcing a non-negative scalar part), and is then mapped to the axis–angle rotation vector $e_{\text{ori}} = \theta\,\mathbf{u} \in \mathbb{R}^3$, where $\theta = 2\,\text{atan2}(\|q_{\text{err},v}\|, q_{\text{err},w})$ and $\mathbf{u} = q_{\text{err},v}/\|q_{\text{err},v}\|$. This 3-dimensional minimal representation removes the redundancy of the unit-quaternion parameterization and is well-defined for $q_{\text{err}} \neq \pm[1,0,0,0]^T$. Since the base is expected to maintain its initial attitude, ~~the orientation error is ideally~~ e_{ori} is kept near zero through the agent~~'~~'s actions. The base ~~'s angular velocity $\omega_{base} \in \mathbb{R}^3$ is also included , as it captures any ongoing rotation~~ angular velocity $\omega_{\text{base}} \in \mathbb{R}^3$ and the base linear velocity $v_{\text{base}} \in \mathbb{R}^3$ are additionally included to capture residual rotation and momentum-coupled translation of the base.

~~Combining these components, the~~ The overall observation vector has dimension ~~$3(e_p) + 3(e_v) + 4(q) + 4(i)$ $3(\dot{q}) + 3(\text{ori}_{\text{base}}) + 3(\omega_{\text{base}})$~~ ~~dimensions. This rich state description provides the agent with all necessary information: how far the end-effector is from target, how the arm is configured/moving, and how the base is reacting~~ $3(e_p) + 3(e_v) + 4(q) + 4(\dot{q}) + 3(\text{ori}_{\text{base}}) + 3(\omega_{\text{base}})$. This state representation contains the variables required by the agent to assess tracking deviation, arm configuration, and base motion.

Action Space~~:~~. The ~~agent's~~ action is a vector of continuous joint torque commands~~for the manipulator's joints~~. At each decision step, the agent outputs ~~$a = [\tau_1\ \tau_2\ \tau_3\ \tau_4]^T$, corresponding to the torque to apply at each of the 4 revolute joints~~ $a = [\tau_1, \tau_2, \tau_3, \tau_4]^T$, with one component per revolute joint of the arm [22], [30]. The~~se~~actions are real-valued and ~~are constrained~~ bounded by the actuator capabilities. ~~We enforce symmetric torque limits : each~~ Symmetric torque limits $|\tau_i| \leq \tau_{\max}$ ~~(for i = 1..4). In the simulation, this is implemented using~~ for $i = 1, \ldots, 4$ are enforced through saturation blocks that clip the commanded torque to the ~~range~~ interval $[-\tau_{\max}, +\tau_{\max}]$. The value of $\tau_{\max}$ is ~~set based on the robot 's design (for example, $\tau_{\max} = 25Nm$ for all joints, though the exact value is not critical as long as it is sufficient~~ selected according to the robot design ($\tau_{\max} = 25\,\text{Nm}$ in this work), provided it is large enough to follow the ~~trajectory). These saturation limits not only model physical actuator limits but also serve as a safety measure to prevent excessive torques~~reference trajectory. The saturation enforces both the actuator limits and an upper bound on the applied torque for safety reasons. The action space is ~~thus continuous~~ therefore $\mathbb{R}^4$, ~~making this a continuous control problem that suits~~ which is compatible with policy gradient and actor-critic ~~RL~~ methods.

The ~~agent's policy (the controller we are learning)~~ policy is a mapping from the 23-dimensional state space to the 4-dimensional action space~~:~~ , $\pi : s \mapsto a$. ~~Internally, this policy is represented~~ parameterized by a neural ~~network~~ whose parameters are adjusted during training to maximize the expected cumulative reward. The ~~control frequency (agent~~

decision frequency) is chosen such that the agent outputs agent issues a new torque command at a fixed interval Δt (e.g., 0.1 s). This Δt is the "agent sampling time" in Simulink and is consistent with the simulation's fixed step to maintain alignment$\Delta t = 0.1\,\mathrm{s}$, which coincides with the fixed simulation step size to ensure consistency between the discrete agent and the continuous plant.

Reward Function Design:. A crucial aspect of the reinforcement learning formulation is the reward signal , which The reward signal guides the agent toward accurate trajectory tracking while minimizing base disturbance and enforcing smooth, safe bounded control actions. At each discrete time step t, the reward r_t is computed as the sum of a progress term and a proximity bonus, minus a weighted cost function:

$$r_t = \frac{1}{C}\left(r_{\text{prog}} + r_{\text{bonus}} - c_t\right), \tag{1}$$

where C is a normalization constant.

To encourage monotonic convergence toward the reference trajectory, a progress reward is defined based on the reduction of the end-effector position error:

$$r_{\text{prog}} = \begin{cases} k_p\left(d_{t-1} - d_t\right), & d_{t-1} > d_t, \\ 0, & \text{otherwise}, \end{cases} \tag{2}$$

where $d_t = \|e_p(t)\|$ is the Euclidean end-effector position error and k_p is a scaling factor. The previous distance d_{t-1} is stored persistently across time steps.

A small shaping bonus is added when the end-effector is close to the reference point:

$$r_{\text{bonus}} = k_b \exp\left(-\left(\frac{d_t}{\sigma}\right)^2\right), \tag{3}$$

which improves local convergence behavior near the desired trajectory.

The instantaneous cost c_t penalizes tracking errors, base motion, and control effort:

$$\begin{aligned} c_t = {} & W_p\|e_p\|^2 + W_v\|e_v\|^2 + W_{\text{ori}}\|e_{\text{ori}}\|^2 + W_{\omega_b}\|\omega_{\text{base}}\|^2 \\ & + W_{v_b}\|v_{\text{base}}\|^2 + W_u\|\tau\|^2 + W_d\|\tau - \tau_{t-1}\|^2, \end{aligned} \tag{4}$$

where e_p and e_v denote the end-effector position and velocity errors, e_{ori} the base orientation error, v_{base} and ω_{base} the linear and angular base velocities, and τ the joint torque vector. The final term penalizes changes in control input to promote smooth actuation.

The weights used in all experiments are summarized in Table 3. High weights are assigned to The physical intuition behind the weight choices is as follows. The end-effector tracking and base orientation preservation, while control effort and position error weight $W_p = 150$ is large because trajectory tracking is the primary task objective. A high weight makes the position error the dominant term of the cost at typical error magnitudes. The velocity error weight $W_v = 25$ is lower than W_p to avoid over-penalizing transient dynamics during fast arm motion, while still discouraging persistent velocity offsets. The base orientation weight $W_{\text{ori}} = 200$ is the largest in the cost function, reflecting that base attitude disturbance is safety-critical in on-orbit operations. Any uncontrolled rotation of the spacecraft base can jeopardize mission success independently of tracking performance. The base angular velocity weight $W_{\omega_b} = 8$ provides a damping-like penalty on ongoing base rotation, while the linear velocity weight $W_{v_b} = 2$ is kept small because, in a free-floating system initialized with zero total linear momentum, base translation is momentum-coupled to the manipulator motion and is expected to remain limited for the considered task configuration. Recall that manipulator motion can induce base translation because the total system center of mass is conserved. The torque magnitude weight $W_u = 0.02$ and torque-rate penalties are kept comparatively small to allow sufficient exploration. weight $W_d = 0.06$ are intentionally small to leave the agent sufficient freedom to explore diverse control actions. The torque-rate penalty is slightly larger to specifically discourage high-frequency chattering in the control signal. The progress scaling factor $k_p = 10$ is chosen so that the progress reward is of comparable magnitude to the cost terms at typical error levels, preventing it from being dominated. The proximity bonus parameters $k_b = 0.05$ and $\sigma = 0.02$ are small by design. They provide a weak local guidance signal near the trajectory without creating a strong local attractor that could trap the policy. Finally, the normalization constant $C = 500$ scales the total reward to a range of approximately $[-1, 0]$ under typical operating conditions, which stabilizes value function learning across all tested algorithms.

At each step, all inputs to the reward function are checked for numerical validity. If any NaN or Inf values are detected, or if the end-effector error exceeds a predefined bound, the episode is immediately terminated and a terminal reward of -1 penalty of $r_{\text{fail}} = -1$ is assigned. This mechanism prevents the agent from exploiting invalid states and ensures numerical robustness during training. value is matched to the reward scale. With the normalization constant $C = 500$, typical step rewards lie approximately in $[-1, 0]$, so $r_{\text{fail}} = -1$ corresponds to the lower end of the one-step reward range. The penalty is therefore large enough to discourage invalid states without dominating the cumulative return over an entire episode. A much smaller penalty (e.g., -0.1) would provide a weak failure signal, whereas a much larger penalty (e.g., -100) could overshadow the remaining reward terms and destabilize value-function learning. Anchoring the failure penalty to the step-reward scale yields a balanced failure signal for policy optimization. All reward components were implemented in a MATLAB Function block within the Simulink environment. The weights were tuned empirically to provide informative gradients while maintaining stable learning behavior.

Training Procedure:. We adopt an episodic training approachAn episodic training scheme is used. Each episode corresponds to a simulation of fixed duration. A finite horizon of H , either a horizon of H seconds or a fixed number of time

steps ~~N. In our case,~~ N, chosen such that one episode covers one ~~episode lasts long enough for the end-effector to attempt following the circular trajectory (e.g., one~~ or two ~~loops of the circle)~~laps of the circular reference. At the start of each episode, the system is reset to an admissible initial condition by a custom reset function. The initial joint angles are set to a feasible configuration ~~(e.g., the arm in a certain pose near the~~ close to the start of the trajectory ~~)~~ and the base is ~~at~~ set to a defined attitude (~~often~~ the home orientation~~and~~ , at rest). All state variables, ~~observer integrators~~integrator states, and logging signals are reset~~. We ensure~~ , and the initial state is ~~within safe bounds~~ verified to lie within the safe set (no collisions or limit violations at ~~t=0~~$t = 0$). The reset function ~~can randomize~~ supports randomization of the starting joint angles ~~within a range or the phase of the trajectory~~ and the trajectory phase to improve generalization~~, though within this study , we kept a consistent start pose for simplicity (~~. In this study a fixed start pose is used,~~ with the end-effector ~~at a specific~~ placed at a defined point on the circle ~~)~~ [22].

~~Once an episode begins~~Within an episode, the agent receives ~~observations~~ an observation at each step and outputs ~~torque actions~~a torque action. The simulation runs until ~~either~~ the time horizon ~~is reached (the episode length H)~~ H is reached or a terminal condition occurs (task ~~completed~~ completion or safety violation). ~~As described, reaching the end of the trajectory successfully or experiencing a failure condition will terminate the episode prematurely. In practice, an episode might end early if~~An early termination is triggered, for example, when the agent causes a collision ~~at 70% of the trajectory, it will receive the penalty~~ before completing the trajectory. The corresponding terminal penalty is applied and the episode ~~stops there~~ends at that instant.

~~We use a~~ The training loop is the standard policy gradient ~~training loop (as~~ procedure provided by MATLAB~~'s RL toolbox): after~~ 's Reinforcement Learning Toolbox. After each episode (or after a ~~set~~ fixed number of steps), the collected state, action, and reward trajectories are used to update the ~~agent's~~ policy (and the value function, ~~if applicable) via~~ where applicable) according to the chosen RL algorithm [42]. The ~~particularities of each algorithm (~~algorithm-specific update rules of PPO, DDPG, ~~etc.)~~ and the remaining methods are handled by their respective ~~implementation, we supply~~ toolbox implementations. This work supplies the experience and ~~let the algorithm update the neural network~~ the reward signal [22]. All agents ~~were~~ are trained for the same number of episodes~~to allow fair comparison. We chose $N_{train} = 1000$ episodes for each agent, which was~~ , $N_{\text{train}} = 1000$, which is sufficient for most algorithms to converge or reach a performance plateau. During training ~~, we monitored metrics such as the cumulative reward per episode ,~~ the cumulative episode reward, the mean end-effector tracking error, and the ~~base motion, averaged over each~~ mean base motion are recorded per episode. A successful training run is characterized by ~~the episode reward increasing (toward 0, since~~ an episode return that increases toward zero (negative costs are used) and by a decrease of the tracking error and base ~~movement decreasing~~ motion over time. ~~We also tracked the~~ The number of episodes ~~that ended prematurely due to safety triggers, aiming for this to go~~ terminated prematurely by the safety monitors is also tracked and expected to decrease to zero as training progresses.

~~For each algorithm, we performed multiple training runs and selected the best-performing policy (based on highest average reward achieved) for final evaluation. We found relatively consistent results~~ To ensure reproducibility, all training runs were initialized using MATLAB's Mersenne Twister random number generator with a fixed seed of zero (`rng(0,'twister')`), applied prior to each training session. For each of the six algorithms, **three independent training runs** were performed under identical conditions, namely the same environment, reward function, episode length, and initial state. This resulted in 18 training runs in total. The policy selection procedure was as follows. After all three runs per algorithm completed, the policy checkpoint achieving the highest mean episodic reward over the final 50 training episodes was identified. This checkpoint was then used exclusively for all subsequent evaluation experiments. The results were relatively consistent across runs for the more stable algorithms (PPO, TRPO), whereas some runs of the less stable methods (e.g.~~DDPG) would diverge, underscoring~~ DDPG) diverged, which underlines the importance of ~~stability~~algorithmic stability in this setting.

Evaluation and Metrics~~:~~. After training, ~~we evaluated each agent's performance~~ each agent's performance was evaluated on $N_{eval} = 30$ independent episodes ~~(with various initial conditions) to gather statistics~~from a fixed initial configuration. Over these evaluation runs, ~~we compute~~ a set of Key Performance Indicators (KPIs) ~~to quantitatively~~ was computed to compare the agents ~~. These KPIs include:~~ quantitatively. For episodes that terminate early due to a safety violation, each KPI is computed over the executed portion of the episode up to the termination step.

The nine KPIs are organized into three groups to aid interpretation. The tracking accuracy group (K_2 and K_3) measures how closely the end-effector follows the reference path. The base stability and safety group covers spacecraft attitude drift (K_4) and constraint violations (K_5 and K_6). The efficiency group covers control smoothness (K_7), residual

TABLE 3: Reward function weights used in all experiments.

Term	Symbol	Value
End-effector position error	W_p	150
End-effector velocity error	W_v	25
Base orientation error	W_{ori}	200
Base angular velocity	W_{ω_b}	8
Base linear velocity	W_{v_b}	2
Joint torque magnitude	W_u	0.02
Torque rate (smoothness)	W_d	0.06
Progress scaling factor	k_p	10
Proximity bonus scaling	k_b	0.05
Proximity width	σ	0.02
Reward normalization constant	C	500

base rotation (K_8), and energy consumption (K_9). K_1 serves as an overall scalar summary of the agent's performance across all objectives, since it is the cumulative reward that jointly penalizes all undesired behaviors.

- ~~K1: Average Episodic Return, the mean~~ **K1** (Average Episodic Return). Mean cumulative reward per episode~~(higher is better , this reflects overall success in the task). $R_i = \sum_{k=1}^{T_i} r_i[k], \quad K1 = \frac{1}{N_{eval}} \sum_{i=1}^{N_{eval}} R_i.$~~ . Higher is better and reflects overall task success.

$$R_i = \sum_{k=1}^{T_i} r_i[k], \qquad K1 = \frac{1}{N_{\text{eval}}} \sum_{i=1}^{N_{\text{eval}}} R_i.$$

- ~~K2: Mean Squared End-Effector Position Error, averaged over the episode (lower is better, measures tracking accuracy). $K2 = \frac{1}{N_{eval}} \sum_{i=1}^{N_{eval}} \frac{1}{T_i} \sum_{k=1}^{T_i} \|e_{p,i}[k]\|_2^2.$~~ **K2** (Mean Squared EE Position Error). Average squared tracking error per episode. Lower is better.

$$K2 = \frac{1}{N_{\text{eval}}} \sum_{i=1}^{N_{\text{eval}}} \frac{1}{T_i} \sum_{k=1}^{T_i} \|e_{p,i}[k]\|_2^2.$$

- ~~K3: Maximum End-Effector Position Error, worst-case deviation during the episode (lower is better~~**K3** (Maximum EE Position Error). ~~$K3 = \max_i \max_k \|e_{p,i}[k]\|_2.$~~ Worst-case tracking deviation. Lower is better.

$$K3 = \max_i \max_k \|e_{p,i}[k]\|_2.$$

- ~~K4: Mean Base Orientation Error, e.g., average quaternion error magnitude indicating how far the base drifted in attitude (lower is better). $K4 = \frac{1}{N_{eval}} \sum_{i=1}^{N_{eval}} \frac{1}{T_i} \sum_{k=1}^{T_i} \|e_{R,i}[k]\|_2.$~~ **K4** (Mean Base Orientation Error). Average attitude drift of the spacecraft base, measured as the magnitude of the axis–angle rotation-vector error e_{ori} (in radians). Lower is better.

$$K4 = \frac{1}{N_{\text{eval}}} \sum_{i=1}^{N_{\text{eval}}} \frac{1}{T_i} \sum_{k=1}^{T_i} \|e_{\text{ori},i}[k]\|_2.$$

- ~~K5: Collision Rate, the fraction of episodes in which~~ **K5** (Collision Rate). Fraction of episodes with at least one collision~~occurred (0 means no collisions, we expect this to be zero if the agent behaves safely). $C_i = \mathbb{I}(\exists k : c_i[k] > 0.5), \quad K5 = \frac{1}{N_{eval}} \sum_{i=1}^{N_{eval}} C_i.$~~ . Zero is ideal.

$$C_i = \mathbb{I}(\exists k : c_i[k] > 0.5), \qquad K5 = \frac{1}{N_{\text{eval}}} \sum_{i=1}^{N_{\text{eval}}} C_i.$$

- ~~K6: Joint Limit Violation Rate (0 is ideal). $K6 = \frac{1}{N_{eval}} \sum_{i=1}^{N_{eval}} \frac{1}{T_i n_J} \sum_{k=1}^{T_i} \sum_{j=1}^{n_J} \mathbb{I}\big(q_{i,j}[k] \notin [q_j^{\min}, q_j^{\max}]\big).$~~ **K6** (Joint Limit Violation Rate). Fraction of time steps across episodes where any joint exceeds its mechanical limit. Zero is ideal.

$$K6 = \frac{1}{N_{\text{eval}}} \sum_{i=1}^{N_{\text{eval}}} \frac{1}{T_i n_J} \sum_{k=1}^{T_i} \sum_{j=1}^{n_J} \mathbb{I}\big(q_{i,j}[k] \notin [q_j^{\min}, q_j^{\max}]\big).$$

- ~~K7: Control Smoothness (Jerk), a measure of actuation smoothness, computed as the second~~ **K7** (Control Smoothness (Jerk)). Second difference of joint torques. Lower values indicate smoother~~control. $\Delta^2 \tau_i[k] = \tau_i[k+2] - 2\tau_i[k+1] + \tau_i[k], \quad K7 = \frac{1}{N_{eval}} \sum_{i=}^{N}$~~ , less chattering actuation.

$$\Delta^2 \tau_i[k] = \tau_i[k+2] - 2\tau_i[k+1] + \tau_i[k],$$
$$J_i = \frac{1}{T_i - 2} \sum_{k=1}^{T_i - 2} \|\Delta^2 \tau_i[k]\|_2,$$
$$K7 = \frac{1}{N_{\text{eval}}} \sum_{i=1}^{N_{\text{eval}}} J_i.$$

- ~~K8: Mean Base Angular Velocity, a measure of residual base rotation (rad/s)~~**K8** (Mean Base Angular Velocity). Residual spacecraft rotation rate during the task (~~lower~~ rad/s). Lower is better, ideally zero~~). $K8 = \frac{1}{N_{eval}} \sum_{i=1}^{N_{eval}} \frac{1}{T_i} \sum_{k=1}^{T_i} \|\omega_{b,i}[k]\|_2.$~~ .

$$K8 = \frac{1}{N_{\text{eval}}} \sum_{i=1}^{N_{\text{eval}}} \frac{1}{T_i} \sum_{k=1}^{T_i} \|\omega_{b,i}[k]\|_2.$$

- ~~K9: Energy Consumption, approximate average energy expended~~ **K9** (Energy Consumption). Average absolute mechanical power integrated over each episode. Lower is better when not at the expense of tracking quality.

$$E_i = \int \sum_{j=1}^{n_J} \big|\tau_{i,j}(t)\, \dot{q}_{i,j}(t)\big|\, dt, \qquad K9 = \frac{1}{N_{\text{eval}}} \sum_{i=1}^{N_{\text{eval}}} E_i.$$

Two KPIs deserve explicit justification in the space mission context. **K7 (Control Smoothness)** captures high-frequency torque oscillations (chattering). In space, such oscillations can accelerate actuator wear, excite structural vibrations that degrade the pointing accuracy of sensitive instruments such as cameras or star trackers, and dissipate energy through rapid back-and-forth motion. Minimizing K_7 therefore supports hardware lifetime and payload pointing requirements. **K9 (Energy Consumption)** reflects the mechanical work performed by the joints~~(e.g., $\int (\tau \cdot \dot{q}) dt$ using an absolute-power proxy). Lower energy usage is better if achieved without sacrificing performance. $E_i = \int \sum_{j=1}^{n_J} |\tau_{i,j}(t) \dot{q}_{i,j}(t)|\, dt, \quad K9 = \frac{1}{N_{eval}} \sum_{i=1}^{N_{eval}} E_i.$~~ , approximated as $\int |\tau \cdot \dot{q}|\, dt$. Onboard power is limited by solar panel area and battery capacity. Energy spent on control actuation reduces the budget available for communication, thermal management, and payload operation. Reducing

K_9 without sacrificing tracking accuracy therefore has operational value for mission planning.

~~Additionally, we consider~~ In addition to the KPIs above, three training-related metrics ~~for comparison: T1:~~ are reported. T_1 is the variance of the episodic reward in the laterstage of training~~(indicating training stability), T2:~~ , an indicator of training stability. T_2 is the variance of the value function or average reward~~(another stability metric), and T3:~~ , a second stability metric. T_3 is the wall-clock training time required ~~to reach~~ for 1000 episodes~~(reflecting~~ , an indicator of sample efficiency and computational load~~). These help evaluate how easily an algorithm learns in this environment.~~ .

~~By evaluating all agents on these common metrics, we obtain an objective comparisonof their performance and characteristics. Section~~ VI ~~presents and discusses these comparative results~~ Applying these metrics to all six agents yields a common basis for comparison. The comparative results are presented and discussed in Section VI.

V. SAFETY MONITORING AND CONSTRAINT ENFORCEMENT

Stability and safety are ~~paramount~~ central requirements in space robotics. A controller for autonomous ~~space servicing must meet physical constraints~~ on-orbit servicing must satisfy the physical constraints of the system and avoid hazardous ~~behaviors, especially when it builds on learned skills expected to handle uncertainties~~states, in particular when the control policy is learned from data and must therefore handle uncertainty. In this work, ~~we integrate~~ run-time safety monitors and ~~constraint enforcement mechanisms~~ constraint-enforcement mechanisms are integrated directly into the RL training and execution environment. ~~These mechanisms serve a dual purpose: they protect~~ They serve two purposes: protecting the simulation from unsafe states during exploration~~and, through penalty-driven learning, shape~~ , and shaping the agent's behavior toward safe operation through penalty-driven learning [22], [24].

~~Momentum Conservation Monitor:~~ **Momentum Conservation Monitor.** In a free-floating system without external forces, both the total linear and the total angular momentum of the robot (base ~~+ arm) should remain constant (and in our case, near zero since initial momentumis zero)~~ [2] ~~. We implemented a Momentum Observer that~~ and arm) are conserved. With zero initial momentum, the total linear momentum remains close to zero throughout the simulation [2]. A momentum observer continuously computes the total linear momentum p_{tot} ~~of the system~~ p_{tot} and checks for its conservation (see Figure 6). ~~This is done by attaching virtual sensors~~ Virtual sensors attached to the base and to ~~each body (base and links) to measure their velocitiesand computing~~ $p_{tot} = \sum_i m_i v_i$ each link provide the body velocities, from which $p_{\mathrm{tot}} = \sum_i m_i v_i$ is computed. The same observer aggregates the link velocities and orientations to track the total angular momentum, which is conserved as well. In an ideal simulation, p_{tot} ~~stays roughly zero at all times (small~~ p_{tot} remains close to zero throughout an episode. Small numerical deviations on the order of ~~10^{-6} Ns may occur due to integration errors). If a significant, systematic change in p_{tot} were detected, it~~ 10^{-6} Ns may arise from time-integration errors. A persistent, systematic drift of p_{tot} would indicate an unintended external force or a modeling error~~(~~, such as an unmodeled constraint~~or bug). During our experiments, p_{tot} remained near 0 (within 10^{-6} Ns)~~ . In the reported experiments, p_{tot} remained within 10^{-6} Ns of zero throughout each episode with the learned policy ~~, confirming physical consistency~~ (see Figure 7)~~. This momentum check provides confidence that the simulation adheres to Newton's laws and the agent interacts with a physical correct process. This leverages the significance of the obtained policy. The Momentum Conservation Monitor~~ , which confirms the physical consistency of the simulation. The momentum observer also serves as a diagnostic tool~~because~~ , as any momentum drift would pause training for model inspection.

~~Collision Monitor and Avoidance: We define a minimum allowable distance between certain~~ **Collision Monitor and Avoidance.** A minimum admissible distance is defined between selected parts of the robot (~~for example,~~ e.g. between the end-effector and the base structure). ~~A Collision Monitor subsystem computes distances between predefined pairs of bodies each step (using the geometry from the URDF and Simscape's collision detection capabilities). If any distance falls below the safe threshold d_{safe} (indicating a potential collision or self-collision is imminent), the system triggers~~ At every time step, a collision monitor computes the distances between the predefined body pairs using the URDF geometry and the Simscape collision-detection facilities. When any distance drops below the safety threshold d_{safe}, a collision event is triggered (see Figure 8) [39]. ~~In our simulation, we set, e.g., $d_{safe} = 5$ cm as the minimum clearance. When triggered, the following occurs: (1)~~A value of $d_{\mathrm{safe}} = 5\,\mathrm{cm}$ is used in this work. Upon activation, three actions are performed, namely (i) a warning flag is raised and logged~~(for analysis), (2)a done signal is sent to terminate the episodeimmediately~~, (ii) a `done` signal terminates the episode, and ~~(3)all joint torque commands from the agent~~ iii) all commanded joint torques are overridden to zero~~to stop the robotmotion. This mechanism effectively acts as an emergency brake, preventing the agent from continuing once a collision is about to happen. The collision monitor was tested by deliberately~~ , stopping the robot. The monitor was verified by intentionally commanding the arm toward the base. It reliably detected the ~~violation of the safety distance~~ threshold violation and stopped the system (see Figure 9). During training~~of the RL agents~~, collisions occurred in early exploratory episodes for some agents, but ~~the monitor caught them and ended those episodes with heavy penalties. As trainingprogressed~~were captured by the monitor and penalized through episode termination. By the end of training, all agents ~~learned to avoid collisions altogether, indeed,~~ avoided collisions in the final evaluation ~~all agents achieved episode without any collision ($K5 = 0$),~~

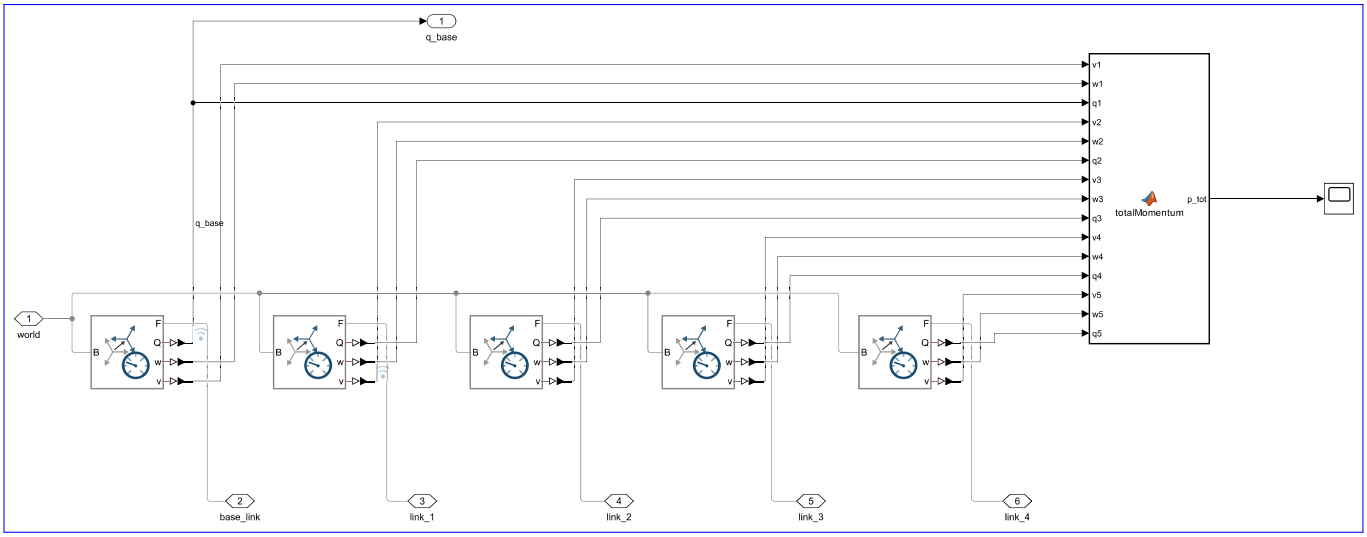


FIGURE 6: Overview of the ~~Impuls~~ Momentum Monitor Subsystem (Momentum Observer) in the Simulink Model `SpaceRobot.slx`
This figure shows the computation of the total linear and angular momentum of the free-floating space robot. The velocities and orientations of the base and all manipulator links are measured and aggregated within a MATLAB function to obtain the system momentum. This monitor is used to analyze momentum conservation and quantify base–manipulator dynamic coupling effects.

~~confirming the efficacy of this safety measure~~($K_5 = 0$ over all 30 evaluation episodes).

~~Joint Limit Enforcement: The robot arm has joint angle limits (e.g., each revolute joint might be limited to ±180° or a smaller range based on mechanical constraints). Violating joint~~ **Joint Limit Enforcement.** Each revolute joint of the arm is subject to mechanical limits. Violating these limits could damage a real robot and ~~often~~ typically indicates an unstable controller. ~~We enforce joint limits~~ The joint limits are enforced at two levels. First, ~~in~~ the Simscape model ~~, we include~~ includes mechanical joint limits with penalty forces~~(~~, i.e. a stiff barrier that ~~resists~~ opposes further motion beyond the limit~~)~~. Second, ~~we add an Assertion block that~~ an assertion block monitors each joint angle. If any joint exceeds its ~~predefined limit~~admissible range, the assertion triggers a violation event ~~. Similarly to the collision case, we handle a joint limit violation by: terminating the episode, applying a large negative~~ that terminates the episode, applies a negative terminal reward, and ~~zeroing out all torques immediately. In fact, the Simulink model is configured such that when the assertion triggers, it directly sets the torque inputs τ to zero in that~~ sets all torques to zero within the same time step~~as a protective measure. This ensures the robot doesn't push further~~ . This prevents further motion into the limit~~or oscillate. In our results, we observed very few limit violations once agents were trained. Among all algorithms,~~ . After training, the joint-limit violation rate is low. PPO and TRPO ~~had a handful of episodes grazing the limits (K6 values of~~ reach the limits in a few percent of evaluation steps (5.1% and 2.1%, respectively), whereas TD3, SAC, PG, and DDPG ~~essentially never hit the limits ($K6 \approx 0$). Notably, PPO 's superior tracking performance came at the cost of occasionally using the full joint range (K6 = 0.0505, i.e. ~5% of episodes, had minor limit violations). Thanks to the joint limit enforcement , these events did not lead to instability: the agent's torques were cut off momentarily when a limit was reached, preventing any damaging behavior,~~ remain within the admissible range ($K_6 \approx 0$). The higher K_6 value of PPO ($K_6 = 0.0505$) coincides with the lowest tracking error and indicates a more aggressive use of the workspace. In all cases, the enforcement mechanism stopped further simulated motion at the limit. The torques were set to zero in the offending step and the episode was marked as failed. ~~Over time, the agent learned to avoid this scenario as well, especially in the optimized PPO version (where K6 dropped to 0~~For the optimized PPO agent (subsection VII-B), no such events are observed in the evaluation episodes ($K_6 = 0$).

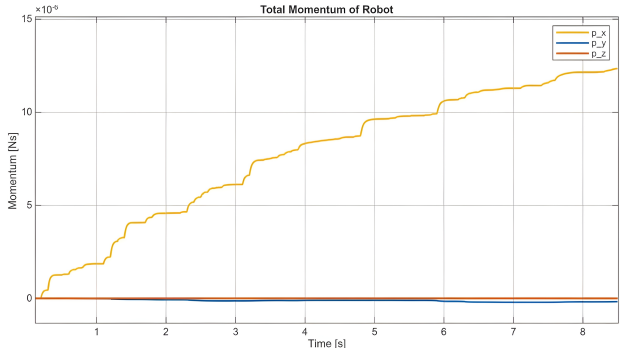


FIGURE 7: Dynamics of the total momentum of the robot.

~~Formal Verification of Torque Constraints:~~ **Formal**

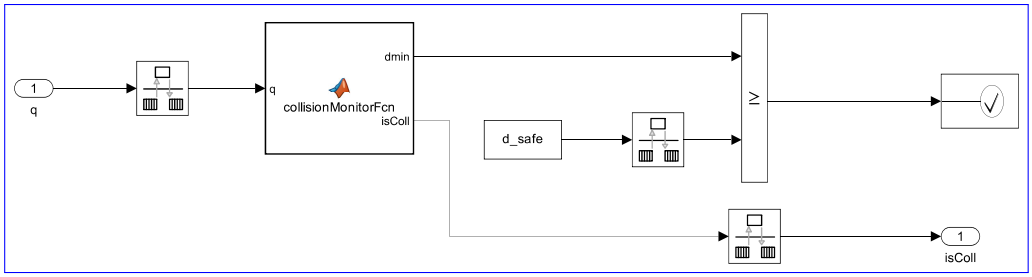
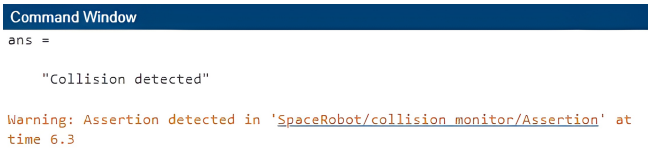
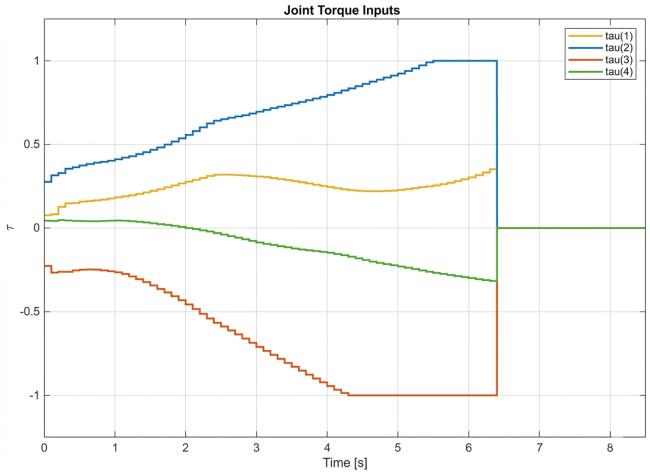


FIGURE 8: Overview of the Collision Monitor Subsystem in the Simulink Model `SpaceRobot.slx`
This figure illustrates the collision monitoring logic used to ensure safe robot operation during training and evaluation. Joint and link configurations are processed to compute the minimum distance to obstacles, which is compared against a predefined safety threshold. A collision flag is generated and used for episode termination or safety-related penalties.



(a) Automatically triggered collision warning



(b) Scope of torque inputs in the event of a collision

FIGURE 9: Representation of collision detection and the associated torque curves.

Verification of Torque Constraints. In addition to ~~enforcing torque saturationduring simulation, we performed~~ runtime saturation, an offline formal verification ~~to prove that~~ was performed to check torque-bound compliance of the trained policy ~~could not command out-of-bounds torques~~block. Using MATLAB ~~'s~~ Simulink Design Verifier (SLDV) in property-proving mode, ~~we analyzed~~ a simplified closed-loop model of the agent ~~controlling the robot's joints~~and the joint torques was analyzed. The property to ~~prove was thatfor all possible agent inputs (observations), each output~~ be checked is that, for any admissible observation input in the analyzed abstraction, each commanded torque τ_i ~~remains within $-\tau_{\max}, +\tau_{\max}$~~satisfies $\tau_i \in [-\tau_{\max}, +\tau_{\max}]$ (see Figure 10). ~~The verification succeeded: SLDV was able to prove that no counterexample exists and the torque limits are always respected by the policy network.~~ SLDV found no counterexample for this property. The corresponding SLDV result is shown in Figure 11 ~~shows the SLDV result confirming the property (no violations found)~~ [23]. This ~~formal assurance is important because it considers all possible states, including ones not seen during simulation, and guarantees the policy cannot output a dangerous torque.~~

~~It's worth noting that directly running formal analysis on~~ result applies to the simplified verification model, including admissible abstract states not visited during simulation. Note that the full Simscape model ~~was not feasible,~~ is not amenable to property proving because of the nonlinear dynamics~~are too complex for the automated theorem proving. To circumvent this, we extracted the relevant part of the model (the control law and saturation logic) into a separate~~ . The verification was performed on a simplified model ~~(Verify_tau.slx) where~~ `Verify_tau.slx`) in which the plant was replaced by a worst-case abstraction~~(essentially, we just needed to check~~ , because only the policy output block ~~). This exemplifies~~ is relevant for the considered property. This illustrates how formal methods can complement simulation-based learning ~~: by verifying critical~~ by checking selected properties on an abstractedmodel of the agent~~, we gain additional confidence in safety beyond empirical testing~~.

~~Limitations of Formal Methods: While the above components significantly enhance safety, we observe that they cover a limited set of properties. They ensure zero collisions, no joint position overshot, and~~ **Limitations of the Formal Methods Component.** The mechanisms above

cover a defined set of monitored properties, namely collision events, joint-limit events, bounded torques, and ~~they monitor physical consistency (e.g., momentum). However, other aspects, such as guaranteeing~~ physical consistency through the momentum check. Other properties, such as task completion within a time ~~or~~ bound or formal stability margins, ~~were not formally~~ are not verified. A complete formal specification of the ~~entire mission(for instance, something like temporal logic specifications of "~~mission, for example temporal-logic specifications such as "the robot shall accomplish ~~X without Y happening") would be far~~ *X* without *Y* happening", would be substantially more complex. ~~Our approach selectively targets the most critical safety concerns. In practice, this proved sufficient to train successful policies without a single catastrophic failure:~~ The present work targets the safety properties that are most relevant for the considered task. Within this scope, the runtime monitors terminated episodes that would have ~~been catastrophic were safely stopped by the monitors (and heavily penalized to teach the agent). As a result, by~~ violated a monitored property and penalized the corresponding actions. By the end of training, the ~~learned agent's behavior was entirely meaningful~~ policy operated within the safety envelope ~~enforced by the formal constraints~~ defined by the monitored constraints in the reported evaluation episodes [25].

In summary, ~~integrating these~~ the formal safety checks

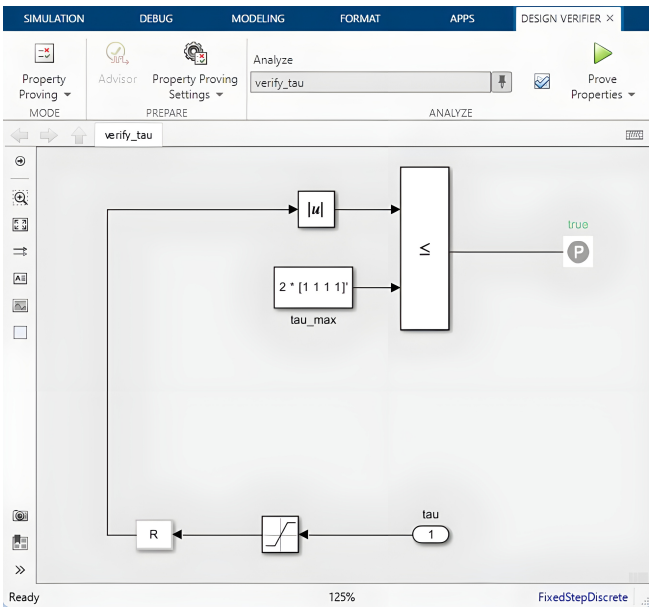


FIGURE 10: Verification model with Simulink Design Verifier `Verify_tau.slx`
This figure depicts the Simulink property model used to formally verify the joint torque constraints. The absolute joint torques are compared against predefined maximum limits using a relational operator. The property is ~~proven to hold for all reachable system states~~checked over the analyzed abstract state space, ~~ensuring~~ supporting bounded control actions in the verification model.

integrated into the learning process ~~was vital. It not only protected the simulation (and a potential future real system) from damage~~ helped prevent unsafe simulated states during the exploratory phase of ~~learning, but it also~~ training and shaped the agent~~'~~'s behavior by penalizing unsafe actions. The agent ~~thus~~ therefore learns with a ~~form of built-in "~~runtime safety shield. ~~" This approach demonstrates a viable path to apply~~ This combination of monitored runtime constraints and penalty-driven learning is one approach to deploying deep RL in safety-critical domains ~~like~~ such as space robotics, ~~where pure~~ in which unconstrained trial-and-error ~~without safeguards would be unacceptable. We next analyze the performance outcomes of the various~~ exploration is not admissible. The next section analyzes the performance of the six RL agents under these conditions.

Computational Overhead and Scalability. The three safety monitors, namely the momentum observer, collision distance checker, and joint-limit assertion, each consist of simple algebraic operations over the robot's link states, scaling as $\mathcal{O}(n_{\text{links}})$ per time step. For the 4-DOF arm considered here, this overhead is negligible compared to the physics simulation itself. End-to-end training times, including the full safety stack, are reported as metric T_3 in Table 5. They range from approximately 8 minutes (PG, PPO) to 38 minutes (TD3) for 1000 episodes on a standard workstation (AMD Ryzen 7 7700, 32 GB RAM). For onboard deployment, only the learned policy network needs to execute in the control loop. The default PPO baseline uses a compact two-hidden-layer network (128 units per layer, $\approx 20\,000$ parameters), while the optimized PPO configuration adopted for the final results (cf. section VII-B) uses a deeper $[256, 256, 128]$ policy with $\approx 1.05 \times 10^5$ parameters, still on the order of 10^5 weights and well below the memory footprint of typical perception networks deployed on space-qualified hardware. At the 10 Hz control rate used here, a single forward pass through this network requires on the order of 2×10^5 multiply-accumulate operations per step, which is compatible with the throughput of modern space-qualified embedded processors (e.g., ARM Cortex-A or LEON-class CPUs with a hardware FPU). The safety monitors can be retained as lightweight parallel watchdog processes, or reduced to the most safety-critical checks (torque saturation, joint-limit assertion) with minimal additional compute. This modular architecture supports scalability to resource-constrained onboard systems.

VI. COMPARISON OF CONTINUOUS RL AGENTS

~~A central question of this study is: Which~~ This section compares six continuous-action RL ~~algorithm is most effective for controlling a free-floating space robot? To answer this, we evaluated six RL~~ algorithms from three ~~major~~ families on the same task and environment [22], [30]. The algorithms ~~compared~~ considered are:

- ~~Policy Gradient (PG): the classic~~ *Policy Gradient (PG)*. The REINFORCE algorithm (Monte Carlo policy gradient), included as a baseline for direct pol-

icy optimization without advanced variance reduction variance-reduction techniques [21], [22].

- Proximal Policy Optimization (PPO): a modern *Proximal Policy Optimization (PPO).* A policy-gradient method that uses clipped probability ratio updates to ensure stable and conservative policy improvements. PPO is known for its balance of stability and performance a clipped probability-ratio surrogate to bound policy updates [20].
- Trust Region Policy Optimization (TRPO): a *Trust Region Policy Optimization (TRPO).* A policy-gradient method that enforces a trust-region constraint (via Kullback–Leibler divergence) on policy updates. TRPO is expected to perform similarly to PPO in terms of stability, albeit with higher computational cost due to solving a constrained optimization each update based on the Kullback–Leibler divergence at each update. The constrained optimization step incurs a higher per-update computational cost than PPO [17].
- Deep Deterministic Policy Gradient (DDPG): an *Deep Deterministic Policy Gradient (DDPG).* An off-policy actor-critic algorithm with a deterministic policy and target networks. DDPG can excel in continuous control by learning a Q-function jointly learns a Q-function and a policy simultaneously, but it is known to be sensitive to hyperparameters and can be unstable and is sensitive to hyperparameter choices [43].
- Twin Delayed DDPG (TD3): an improved *Twin Delayed DDPG (TD3).* A variant of DDPG that addresses function approximation issues by using uses two critic networks to mitigate reduce overestimation bias and by delaying policy updates . TD3 generally offers more stable learning than standard DDPG delays policy updates relative to value updates [18].
- Soft Actor-Critic (SAC): an *Soft Actor-Critic (SAC).* An off-policy actor-critic algorithm that learns with a stochastic policy and includes an entropy term in the reward to encourage exploration. SAC aims to achieve both high reward and high entropy , making the agent's behavior robust and exploratory. It often achieves very good performance but can be computationally heavier due to sampling and entropy calculations objective. This entropy regularization promotes exploration but increases the per-step computational cost relative to deterministic methods [19].

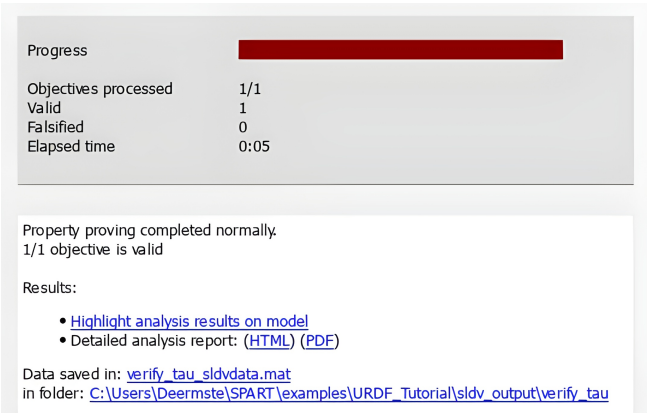


FIGURE 11: Result of the Simulink Design Verifier
This figure shows the result of a formal property verification performed on the torque constraint model. The verification confirms that the specified safety property is satisfied for all analyzed execution paths. This analysis provides formal guarantees on control torque limits model-based evidence for torque-limit compliance beyond empirical simulation-based validation.

TABLE 4: Classification of the evaluated reinforcement learning algorithms.

Algorithm	Policy Type
Policy Gradient (PG)	On-policy
Proximal Policy Optimization (PPO)	On-policy
Trust Region Policy Optimization (TRPO)	On-policy
Deep Deterministic Policy Gradient (DDPG)	Off-policy
Twin Delayed DDPG (TD3)	Off-policy
Soft Actor-Critic (SAC)	Off-policy

These algorithms cover a spectrum from

The six algorithms cover the on-policyto /off-policy , from deterministicto stochastic , and from simpler to more complex and deterministic/stochastic axes, as well as a range of update rules. Based on prior knowledge and the nature of our problem, we hypothesized that algorithms with more robust, stable Given the strong base–manipulator coupling and the safety constraints, methods with constrained or clipped policy updates (like PPOand TRPO) would likely outperform others in this highly coupled, continuous control scenario. Off-policy PPO, TRPO) are expected to provide more stable training than algorithms without such constraints (PG, DDPG). The off-policy actor-critic methods (DDPG, TD3, SAC) might reach good performance as well, but could suffer from tuning sensitivity and stability issues given the complex dynamics and safety constraints. Indeed, DDPG and basic PG were expected to be the least reliable due to known tendencies for instability without additional regularizationmay achieve comparable performance but are typically more sensitive to hyperparameter selection.

On-policy methods (PG, PPO, TRPO) update their policies using only data from the policy only from data generated by the current policy, leading which typically leads to stable but less sample-efficient learning. Off-policy methods (DDPG, TD3, SAC) reuse experience from stored in a replay buffer, improving which improves sample efficiency at the cost of higher sensitivity to hyperparameters [22], [30]. For free-floating space robot control , where space-robot control with strong nonlinear coupling and safety constraintsdemand reliable convergence, , the constrained-update on-policy methods are expected to be advantageousprovide more reliable convergence.

Training Setup:. All experiments were conducted performed on a workstation equipped with an AMD Ryzen

7 7700 processor and 32 GB of RAM, using MATLAB R2025b with the Reinforcement Learning Toolbox and Simscape Multibody. ~~We used the~~ The same network architecture (~~initially~~ two hidden layers of 128 units each for ~~policy and for value function~~the policy and, where applicable~~) and same reward structure for all . Hyperparameters (learning rate, discount factor~~ , for the value function) and the same reward function were used for all agents. The discount factor was fixed at $\gamma = 0.99$~~, etc.) were initially set to default reasonable values and kept consistent across agents as much as possible. All agents were trained using a learning rate of $\alpha = 1 \times 10^{-3}$~~ . The learning rate for both actor and critic networks ~~,~~ was $\alpha = 1 \times 10^{-3}$ unless stated otherwise. ~~Hence, any~~ With these unified settings, observed performance differences can be attributed primarily to the algorithmic differences ~~, not~~ rather than to unequal tuning. ~~Later we will tune PPO further, but for~~ PPO was further tuned in a subsequent step. In the main comparison, PPO uses ~~a default configurationas well~~its default configuration. Each agent ~~'s training run of~~ was trained for 1000 episodes~~yielded a final policy which we~~ , and the resulting policy was then evaluated.

Performance Metrics~~:~~. ~~We evaluated the agents~~ The agents were evaluated on the KPIs ~~described earlier (tracking error, base stability, etc.)~~ introduced in Section IV by running 30 ~~test episodes for each.~~ evaluation episodes per agent. Table 5 ~~summarizes~~ reports a subset of these metrics ~~, highlighting the differences between~~ for the six agents.

The results in Table 5 ~~reveal clear patterns across the six agents. Regarding~~ display the following patterns. With respect to training stability, TRPO and PPO ~~exhibited the most stable learning (T1 values of 0.61 and 1.78~~yield the smallest late-training reward variance ($T_1 = 0.61$ and $T_1 = 1.78$, respectively), whereas DDPG ~~showed highly erratic behavior (T1 = 64.5). PPO was also the fastest to train (T3 ≈ 8 min), while~~ shows the largest ($T_1 = 64.5$). The training time T_3 is shortest for PPO ($\approx 8\,\text{min}$). The off-policy methods ~~required substantially longer~~ are slower (TD3 ~~≈ 38 min, SAC ≈ 35 min~~≈ 38 min, SAC ≈ 35 min).

~~In terms of~~ For trajectory tracking, PPO ~~achieved~~ attains the lowest mean squared error (~~K2 = 0.0257), significantly outperforming the runner-up TRPO (0.0680~~$K_2 = 0.0257$), lower than that of TRPO ($K_2 = 0.0680$). DDPG ~~achieved~~ attains the lowest peak error (~~K3 = 0.4758) in its best runs but was highly inconsistent~~$K_3 = 0.4758$) but with large variability across runs. PG, SAC, and TD3 ~~all exhibited substantially~~ yield larger tracking errors~~, with PG's mean error roughly~~ . The mean error of PG is approximately six times that of PPO.

For ~~base motion~~ base-motion minimization, PPO and TRPO ~~both kept the base orientation error small (K4 of 0.0667 and 0.0694~~yield comparable base-orientation errors ($K_4 = 0.0667$ and $K_4 = 0.0694$), while DDPG ~~allowed the base to drift considerably (K4 = 0.2682). Interestingly,~~ produces a larger drift ($K_4 = 0.2682$). TD3 ~~achieved very~~ yields a low base angular velocity (~~K8 ≈ 0.02 rad/s) by adopting an overly conservative strategy that sacrificed tracking accuracy~~$K_8 \approx 0.02\,\text{rad/s}$) at the cost of degraded tracking accuracy, consistent with a conservative policy that suppresses actuation. All agents ~~successfully avoided collisions (K5 = 0~~avoid collisions ($K_5 = 0$). PPO and TRPO ~~occasionally approached joint limits (K6 of 5.1% and 2.1%~~approach the joint limits in a small fraction of the steps ($K_6 = 5.1\%$ and $K_6 = 2.1\%$, respectively), ~~indicating more aggressive~~ corresponding to a wider use of the ~~workspace for better tracking~~joint range.

PPO ~~also produced~~ produces the smoothest control signals (~~K7 = 0.6812), while~~ $K_7 = 0.6812$). The highest jerk values are observed for TD3 and DDPG ~~generated the jerkiest outputs (≈ 0.8–1.0). Energy consumption~~ (~~K9) was~~ $K_7 \approx 0.8$–1.0). The energy consumption K_9 is moderate for PPO (~~≈ 6.33~~≈ 6.33) and lowest for TRPO (~~≈ 3.76); the~~ ≈ 3.76). The very low values for DDPG and TD3 (~~< 1.0) reflect insufficient actuation rather than~~ $K_9 < 1.0$) coincide with insufficient actuation and should not be interpreted as energy efficiency. Qualitatively, PPO ~~produced the most competent behavior: the arm moved accurately along the circle with minimal base rotation. TRPO performed similarly but with slightly~~ traces the circular reference with low base disturbance. TRPO produces a similar profile with a slower response. SAC ~~tended to oscillate around the path,~~ oscillates around the reference. TD3 ~~adopted an overly conservative strategy sacrificing tracking accuracy, DDPG was~~ adopts a conservative strategy that reduces tracking accuracy. DDPG is inconsistent across runs~~(explaining its high T1), and PG never achieved~~ , in line with the large T_1. PG does not reach reliable tracking within 1000 episodes.

~~**Overall Winner, PPO:**~~**Summary of the comparison.** Aggregating ~~all the criteria , PPO stands out as the best-performing agent. As summarized succinctly in the report~~the criteria above:

- PPO offers the best combination of training stability (low ~~T1/T2~~T_1/T_2), training ~~speed (short T3~~time (T_3), tracking accuracy (low ~~K2/K3~~K_2/K_3), and control smoothness (low ~~K7), with a moderate joint limit~~ K_7), at the cost of a moderate joint-limit violation rate.
- TRPO is ~~a close second, described as similarly stable and precise but requiring~~ close to PPO on most metrics, but requires more computation time and still ~~incurring slight limit~~ incurs occasional joint-limit violations.
- The off-policy methods (TD3, SAC) ~~did not achieve better performance and were significantly~~ do not improve on PPO under unified conditions and are more computation-intensive, with higher tracking errors.
- ~~Finally,~~ PG and DDPG ~~were less robust both in training and in achieved performance, showing~~ show the largest fluctuations and ~~poorest metrics~~the weakest performance on most KPIs.

These ~~findings validate our hypothesis that~~ results are consistent with the on-policy~~methods with robust update rules are best suited for this highly coupled, safety-critical~~

TABLE 5: Selected performance metrics for each RL agent (best ~~values~~ value in each row in bold). PPO ~~excels in most categories, achieving~~ attains the lowest tracking errors (K2, K3), ~~best base stability~~ the lowest base-orientation error (K4), and ~~smoothest control~~ the lowest jerk (K7), ~~while maintaining short~~ with the shortest training time (T3). TRPO is ~~a~~ close ~~second in many~~ to PPO on several metrics but requires more training time. ~~Off-policy~~ The off-policy methods (TD3, SAC, DDPG) show larger errors or ~~instabilities~~ a larger late-training reward variance (e.g., high T1 for DDPG)~~despite some strengths (~~. DDPG ~~'s low max~~ attains the lowest maximum error (K3). ~~KPI data from evaluation of~~ Values are averaged over 30 evaluation episodes per agent.

KPI (unit)	TRPO	TD3	SAC	PPO	PG	DDPG
K2: Mean Squared EE position error	0.0680	0.1914	0.1908	**0.0257**	0.1638	0.0937
K3: Max EE error	0.6783	0.5341	0.7766	0.4833	1.0654	**0.4758**
K4: Mean base orientation error	0.0694	0.0868	0.1687	**0.0667**	0.1302	0.2682
K6: Joint limit violation rate	0.0214 (2.1%)	0	0	0.0505 (5.1%)	0	0
K7: Torque smoothness (norm. jerk)	0.7498	0.9981	0.7536	**0.6812**	0.7125	0.8118
T1: Reward std. (late training)	**0.6144**	8.5168	9.6907	1.7756	13.1970	64.4764
T3: Training time (1000 eps)	∼13 min	∼38 min	∼35 min	**∼8 min**	∼8 min	∼24 min

~~control problem~~/off-policy distinction in Table 4: under the strong dynamic coupling and the safety constraints of the considered task, the constrained-update on-policy methods are the most reliable. Based on these results, ~~we selected PPO for further optimization~~PPO is selected for further tuning.

~~Based on these results, we selected PPO as the reference agentfor further development and optimization. The next section discusses how we improved PPO 's performance even further through targeted optimizations~~The next section reports the architectural and hyperparameter modifications applied to PPO and their effect on the KPIs.

Statistical Significance of Agent Comparisons. To substantiate the qualitative ranking above with statistical evidence, non-parametric hypothesis tests were applied to the 30 evaluation episodes per agent. First, a Friedman test confirmed that at least one agent differs significantly from the others for every KPI (all $p < 10^{-17}$), justifying pairwise comparisons. Subsequently, pairwise Wilcoxon signed-rank tests were conducted between PPO (baseline) and each competing agent, with Holm correction applied to control the family-wise error rate across multiple comparisons. Table 6 reports the Holm-corrected p-values and the median differences (positive values indicate PPO is superior for a minimization objective, negative for a maximization objective).

The statistical analysis confirms and refines the conclusions drawn from the summary metrics. For episodic return (K_1), TRPO and TD3 achieve marginally higher returns than PPO ($\Delta\tilde{x} = +7.55$ and $+8.90$, respectively), yet both converge to a highly conservative strategy. TD3 in particular yields $K_7 \approx 0.998$ and $K_9 < 1$, indicating near-zero actuation rather than effective tracking, which is reflected in its substantially larger tracking errors (K_3 and K_4 significantly worse). SAC, PG, and DDPG produce significantly lower returns than PPO ($p < 0.001$ each), with DDPG showing the largest deficit ($\Delta\tilde{x} = -102$).

For tracking accuracy (K_2, K_3), PPO is statistically superior to TRPO, TD3, and SAC (all $p < 0.001$), while differences versus PG and DDPG do not reach significance at the $\alpha = 0.05$ level after Holm correction, a consequence of high variability in those agents. The effect size for SAC is particularly pronounced (K_2, $\Delta\tilde{x} = +0.908$), confirming that SAC's oscillatory behavior causes consistently large position errors.

Regarding base stability and safety (K4–K6), PPO significantly outperforms TD3, SAC, PG, and DDPG in base orientation error (K_4, all $p < 0.01$), while the difference against TRPO is not significant ($p = 0.926$), indicating comparable stability for on-policy methods. For the collision-rate KPI (K_5), all evaluated agents achieve $K_5 = 0$ over the 30 evaluation episodes. The collision metric therefore does not distinguish the agents in this evaluation and is reported descriptively rather than used for ranking.

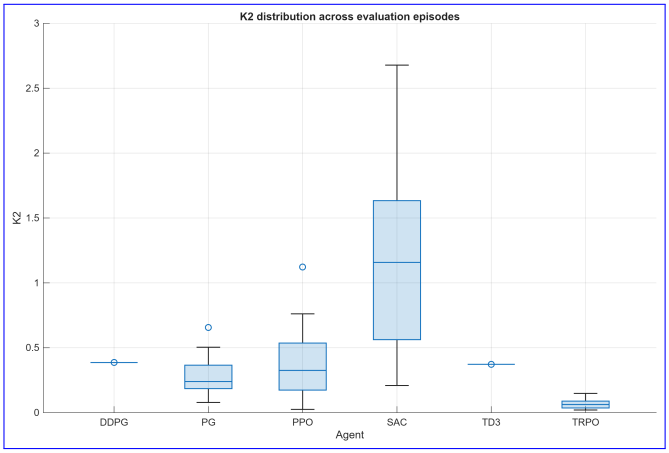
For control quality and efficiency (K7–K9), PPO produces significantly smoother control signals than all competing agents (K_7, all $p < 0.001$), with DDPG and TD3 showing the largest jerk deficits ($\Delta\tilde{x} = +0.724$ and $+0.731$). Energy consumption (K_9) is significantly lower for TRPO, TD3, and DDPG compared to PPO. However, as noted for K_1, these low values reflect inadequate actuation rather than genuine efficiency, and must be interpreted alongside the accompanying tracking errors.

Overall, the statistical tests corroborate that PPO offers the best balance across all KPI dimensions. It outperforms the field on tracking accuracy and control smoothness, maintains competitive base stability, and does so with statistically verifiable consistency across 30 independent evaluation episodes. The distribution of four key KPIs across all 30 evaluation episodes is visualized as boxplots in Figure 12 (R2.13), confirming the consistency of the reported medians and revealing the spread and outlier structure for each agent. Figure 13 (R2.12) shows the corresponding 95% bootstrap confidence intervals, providing a quantitative measure of estimation uncertainty per agent and KPI.

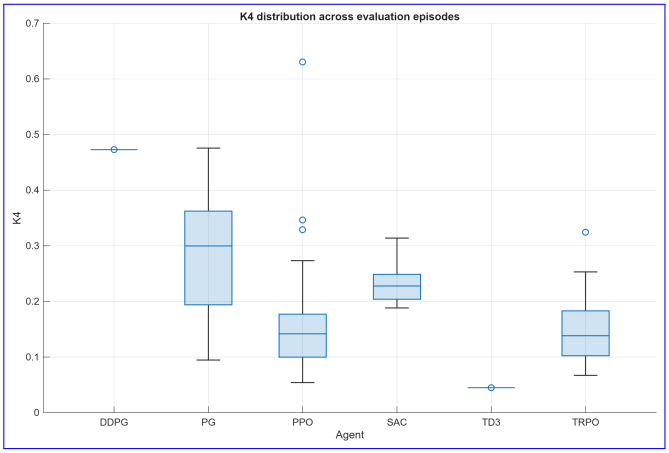
VII. OPTIMIZATION OF THE PPO AGENT

TABLE 6: Pairwise Wilcoxon signed-rank tests of each agent vs. PPO (baseline), Holm-corrected. Each cell shows $\Delta\tilde{x}$ (median difference, agent−PPO) with significance superscript: $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$, $^{\text{n.s.}}p \geq 0.05$. For K1, positive $\Delta\tilde{x}$ means the agent achieves a higher (better) return. For K2–K9, negative $\Delta\tilde{x}$ means lower (better) error/violation/jerk/energy values. K5 is the collision rate (0 = no collisions). All evaluated agents achieved K5 = 0.

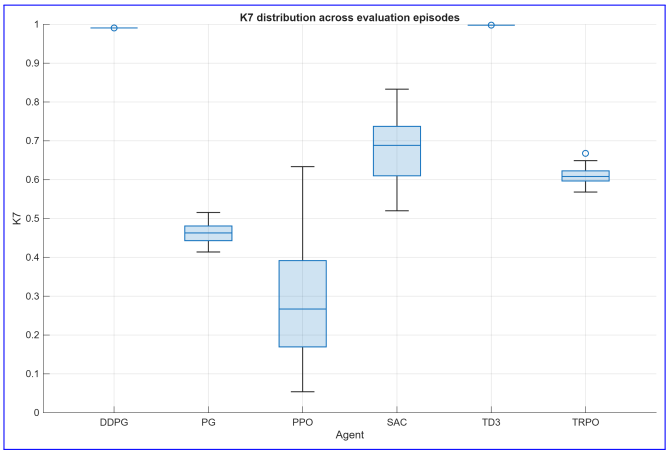
Agent	K1	K2	K3	K4	K5	K6	K7	K8	K9
TRPO	$+7.55^{*}$	-0.277^{***}	-0.526^{***}	$+0.003^{\text{n.s.}}$	$0^{\text{n.s.}}$	-0.137^{***}	$+0.362^{***}$	-0.041^{***}	-9.38^{***}
TD3	$+8.90^{***}$	$+0.047^{\text{n.s.}}$	-0.276^{***}	-0.097^{***}	$0^{\text{n.s.}}$	-0.156^{***}	$+0.731^{***}$	-0.095^{***}	-12.91^{***}
SAC	-33.79^{***}	$+0.908^{***}$	$+0.981^{***}$	$+0.093^{**}$	$0^{\text{n.s.}}$	-0.059^{***}	$+0.404^{***}$	$+0.032^{***}$	$+2.44^{*}$
PG	-24.46^{*}	$-0.051^{\text{n.s.}}$	$-0.132^{\text{n.s.}}$	$+0.158^{**}$	$0^{\text{n.s.}}$	-0.156^{***}	$+0.199^{***}$	$0.000^{\text{n.s.}}$	-4.05^{***}
DDPG	-102.02^{***}	$+0.060^{\text{n.s.}}$	$-0.095^{\text{n.s.}}$	$+0.331^{***}$	$0^{\text{n.s.}}$	-0.153^{***}	$+0.724^{***}$	$-0.006^{\text{n.s.}}$	-12.17^{***}



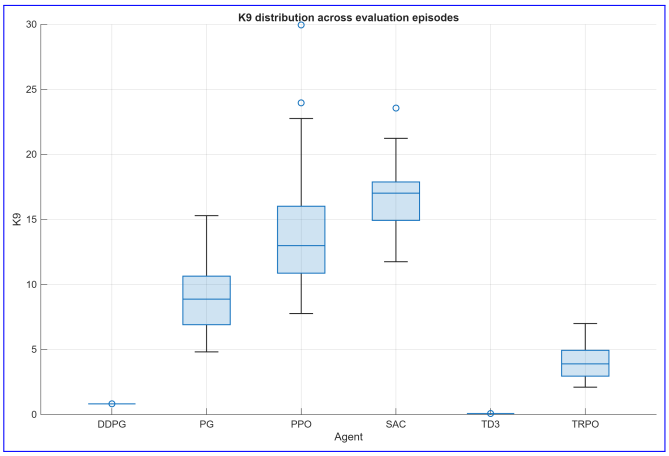
(a) K2: Mean Squared EE Position Error



(b) K4: Mean Base Orientation Error



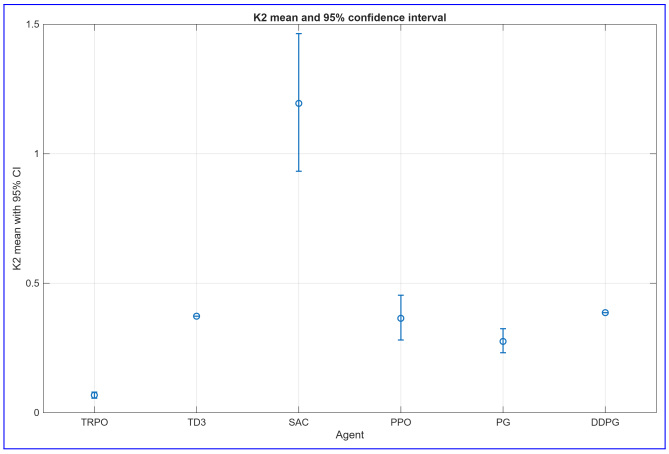
(c) K7: Control Smoothness (Norm. Jerk)



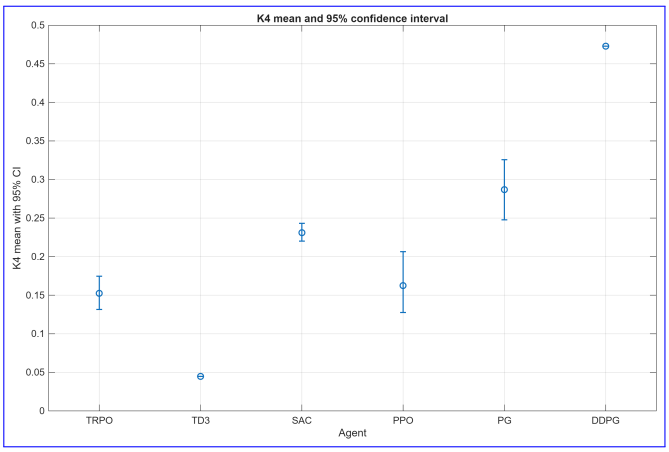
(d) K9: Energy Consumption

FIGURE 12: Boxplots of selected KPI distributions over 30 evaluation episodes per agent. Each box spans the interquartile range (IQR), the center line marks the median, the whiskers extend to 1.5 × IQR, and circles denote outliers. PPO consistently achieves low tracking error (K_2) and smooth control (K_7) with small spread, while PG and SAC show high variability. The low K_9 values for TD3 and DDPG reflect insufficient actuation rather than energy efficiency.

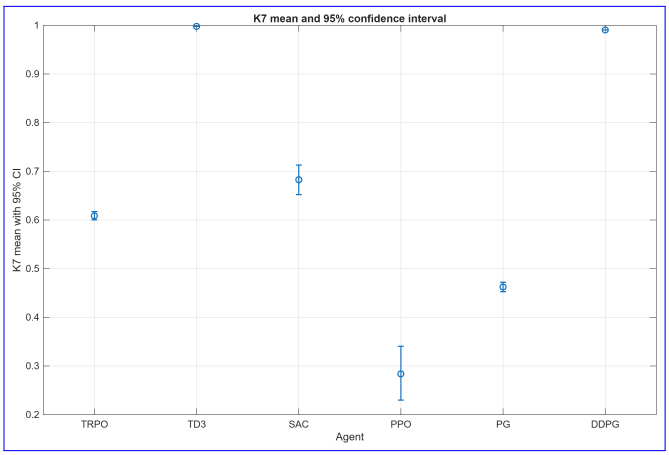
~~Having identified PPO as the most promising algorithm, we focused on enhancing its performance on path tracking task of the space robot.~~ Based on the comparison in Section VI, PPO is selected for further tuning. The PPO agent ~~we compared above was a "default " PPO configuration (as~~ used so far corresponds to the default configuration provided by MATLAB~~'s RL Toolbox example) with minimal tuning [42]. While it already outperformed other agents, we hypothesized there was room to improve in terms of~~ 's Reinforcement Learning Toolbox example [42], with minimal manual adjustment. The following modifications target three objectives: faster convergence, ~~higher precision,~~



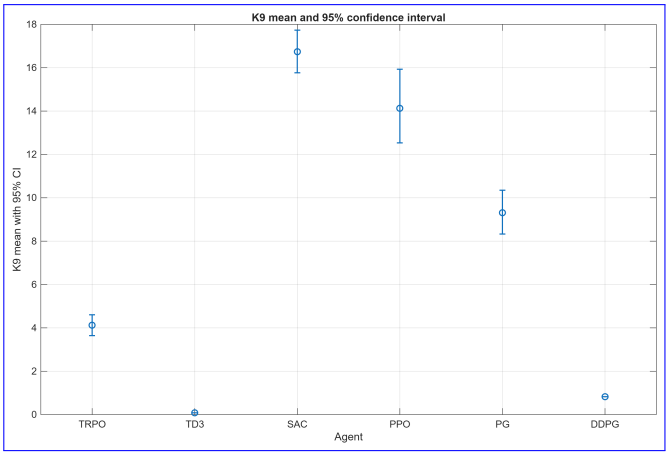
(a) K2: Mean Squared EE Position Error



(b) K4: Mean Base Orientation Error



(c) K7: Control Smoothness (Norm. Jerk)

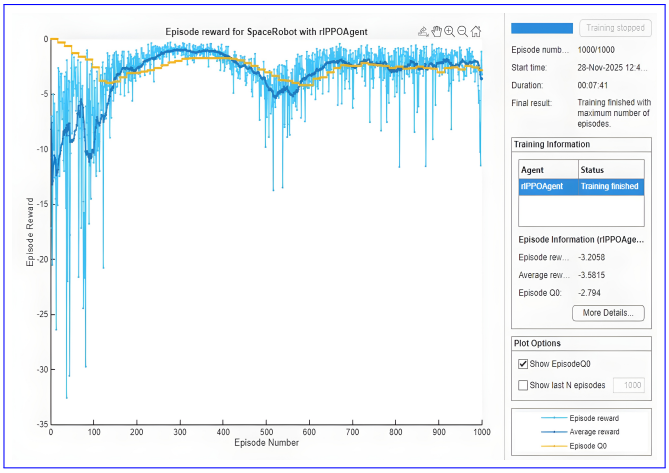


(d) K9: Energy Consumption

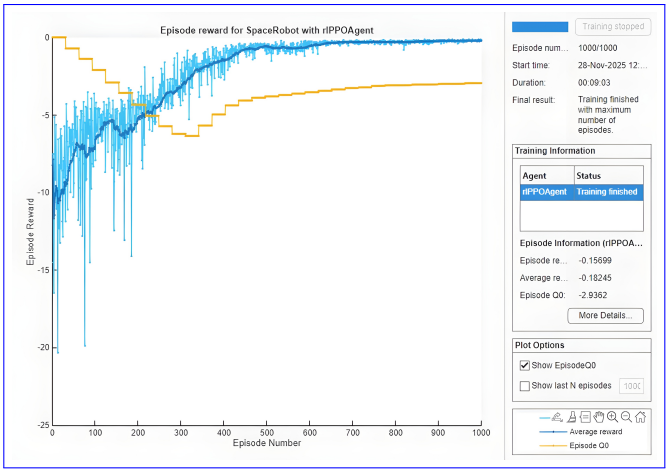
FIGURE 13: 95% bootstrap confidence intervals for selected KPIs across all agents (30 evaluation episodes per agent). Narrow intervals for TD3 and DDPG reflect near-constant behavior (policy collapse), whereas PPO's intervals are tight despite exploring the full workspace. TRPO and PPO show overlapping intervals for K_4 (base orientation), consistent with the non-significant Wilcoxon result ($p = 0.926$).

~~and eliminating the remaining constraint violations. We therefore undertook a systematic optimization of PPO, involving~~ lower tracking and base-orientation errors, and avoidance of the remaining joint-limit violations. The tuning involves both network architecture ~~modifications and hyperparameter tuning~~and hyperparameters.

~~Network Architecture Improvements: By default , the PPO agent used~~ **Network architecture.** The default PPO agent uses two hidden layers of 128 ~~neurons each (~~units each with ReLU activations ~~)~~ for both the policy and ~~value function networks. We experimented~~ the value network. Variants with deeper and wider networks ~~, as well as~~ and with alternative activation functions ~~, to better capture the dynamics of the problem. Our optimized architecture ultimately used 3 hidden layers~~ were evaluated. The selected architecture uses three hidden layers of sizes $[256, 256, 128]$ for the policy network ~~with layer sizes 256, 256, 128and 2 hidden layers~~ and two hidden layers of sizes $[256, 128]$ for the value network~~with sizes 256, 128. We found that the~~ . The additional depth and capacity ~~in~~ of the policy network ~~helped the agent learn the intricate relationship between arm motions and base reactions (especially given~~ reduce the tracking error and the base-orientation error (cf. Table 7) for the 23-dimensional state input ~~)~~ [42]~~. We also introduced layer normalization in~~ [42]. Layer normalization is applied to the first layer to ~~help stabilize learning, given~~ compensate for the different scales of the input features (~~joint angles vs. orientation quaternions, etc.~~e.g., joint angles versus orientation quaternions). The activation function ~~remained ReLU, which was sufficient. These architecture changes were implemented by explicitly defining the neural network layers instead of using the auto-generated default network. In summary, the optimized policy network was larger and potentially more expressive, which we believe allowed PPO to find a control strategy which is better tuned.~~ is kept as ReLU. The networks are defined explicitly rather than through the toolbox default



(a) PPO (Default)



(b) PPO (Optimized)

FIGURE 14: Comparison of episodic rewards for default and optimized PPO agents. (a) Episode reward evolution during training using the default PPO hyperparameters. (b) Episode reward evolution using the optimized PPO configuration. The optimized configuration ~~exhibits~~ shows faster convergence and a higher average ~~episode rewards, indicating improved learning efficiency and policy performance~~episodic return than the default configuration.

constructor.

~~Hyperparameter Tuning: We systematically varied key PPO hyperparameters and evaluated their impact on training curves (learning speed and stability):~~ **Hyperparameter Tuning.** The optimization followed a coordinate-wise grid search. Each hyperparameter was swept across a discrete set of candidate values while all others were held at their current best, and the configuration yielding the highest mean episodic reward over the final 50 training episodes (out of $N_{\text{train}} = 1000$) was selected. In total, 16 training runs of 1000 episodes each were conducted during this tuning phase. The swept parameters, their candidate values, and the selected values are as follows:

- ~~The learning rate was tuned: the default was 1e-3. We found that a slightly lower learning rate (e.g. 5e-4) improved stability (preventing overshooting in policy updates) without slowing convergence.~~ **Actor learning rate.** Candidates $\{1\times10^{-3},\ 5\times10^{-4}\}$, selected 5×10^{-4}. The lower rate reduced overshooting in policy gradient updates, improving convergence stability.
- ~~The mini-batch size for stochastic gradient descent updates was increased. By default, PPO might use a batch of 64 from each trajectory. We increased this to 256 to provide more stable gradient estimates per update, which helped reduce the variance in training.~~
- ~~The clip ratio (ϵ) for PPO's surrogate objective was adjusted. The default clip value 0.2 worked, but we tried slightly tighter (0.15) and looser (0.3) clips [20]. A tighter clip (ϵ = 0.1–0.15) made updates more conservative. This reduced occasional policy overshoots at the cost of slower improvement. We settled on $\epsilon \approx 0.2$ as it was already near-optimal, but this process confirmed that too high a clip would harm stability. The clip factor ϵ in PPO limits the magnitude of policy updates by constraining the probability ratio between the new and the old policy . This prevents excessively large policy updates that could destabilize training, particularly in highly nonlinear control tasks. In safety-critical scenarios such as free-floating space robot control, clipping contributes significantly to stable and reliable convergence [20], [42].~~ **Critic learning rate.** Candidates $\{1\times10^{-3},\ 5\times10^{-4}\}$, selected 1×10^{-3}. A higher critic learning rate allowed the value function to track the evolving policy more closely, providing more accurate advantage estimates.
- ~~Entropy regularization weight was tuned. PPO can include an entropy term to encourage exploration. We increased the entropy coefficient modestly (from 0 to 0.01) to avoid~~ **Entropy loss weight.** Candidates $\{1\times10^{-2},\ 1\times10^{-3}\}$, selected 1×10^{-3}. A small entropy bonus discouraged premature convergence to a suboptimal deterministic policy ~~. This indeed improved the exploration in early training, helping the agent discover better strategies (like moving joints in opposite directions to~~ and encouraged the agent to discover strategies such as coordinated joint motions that cancel base momentum~~)~~, without causing aimless exploration.
- ~~Discount factor (γ) and GAE (Generalized Advantage Estimation) parameter (λ) were left at standard values ($\gamma = 0.99$, $\lambda = 0.95$) after confirming that they already provided a good bias-variance trade-off for advantage estimation~~ **Experience horizon.** Candidates $\{512,\ 1024,\ 2048\}$, selected 1024. A moderate rollout horizon provided sufficient trajectory information per

update while keeping memory requirements and update latency manageable.

- ~~We also adjusted the target KL-divergence for PPO's stopping criteria (if used) to ensure the policy updates didn't diverge. We monitored the KL and it remained within acceptable range with our chosen settings.~~ **Mini-batch size.** Candidates $\{128, 256\}$, selected 256. Larger batches provided more stable gradient estimates per update, reducing training variance.

~~Through a grid search over these parameters (one at a time and some combined), we converged on a set~~ This process yielded a configuration that provided a more stable and higher-performing PPO agent. Notably, the ~~optimized agent is tailored to this specific task and reward function, for example, the entropy bonus was chosen to the point where it encourages exploration of different arm motions, but not so high that the agent keeps thrashing the arm aimlessly~~asymmetric learning rates, a slower actor (5×10^{-4}) paired with a faster critic (1×10^{-3}), reflect the different roles of the two networks. The value function must adapt quickly to policy changes, while conservative actor updates prevent destabilizing the learned control strategy.

A. CIRCULAR TRAJECTORY

~~Training Outcome: The effects of these optimizations were evident~~ **Training outcome.** The effect of the modifications can be seen in the training ~~learning~~ curves (see Figure 14). The default PPO agent~~'s learning curve initially rose, but then plateaued at a fairly negative reward value (~-2 per episode) and exhibited high variance even after many episodes. This plateau suggested it reached~~ 's episodic return increases initially and then plateaus around -2 per episode with high variance over the remaining training, indicating convergence to a suboptimal policy~~and struggled to improve further~~. In contrast, the optimized PPO agent~~'s learning curve kept rising steadily and converged to a much higher average reward (less negative,~~ 's return continues to increase and converges to a higher value (closer to zero) with ~~significantly~~ reduced variance. Table 7 reports the KPI averages over ~~$N_{eval} = 30$~~ $N_{\text{eval}} = 30$ evaluation episodes for PPO with the default ~~configuration~~ and the optimized ~~configuration. Specifically, where the default PPO leveled off around -2.0 cumulative reward, the optimized PPO approached -0.4 on average, a dramatic improvement. The variability (shaded region in the reward plot) shrank, indicating the agent's behavior became consistently optimal across training episodes. Quantitatively, the mean episodic reward K1 improved from -2.23 (default PPO) to -0.43 (optimized PPO). This roughly fivefold increase in reward corresponds to the agent accomplishing much more of the task (since zero would be perfect tracking with no penalties). The lower variance also hints that the agent not only improved its average performance but did so reliably without frequent failures~~configurations. The mean episodic return K_1 increases from -2.23 for the default PPO to -0.43 for the optimized PPO, an increase by a factor of approximately five. The reduced variance indicates that the improvement is consistent across episodes rather than driven by a few outliers.

~~Beyond the reward improvement , the optimizations translate directly to task~~ The reward improvement translates directly into task-level performance. Figure 15 shows the end-effector path in the ~~XY-plane~~ XY plane: the default PPO deviates ~~significantly~~ from the reference circle and exhibits ~~jagged oscillations, whereas~~ oscillations, while the optimized PPO ~~traces the referencealmost exactly~~closely follows the reference. The mean squared tracking error ~~K2 dropped from 0.0257 to 0.0083 (a 67.7% reduction~~ K_2 decreases from 0.0257 to 0.0083 (a reduction of 67.7%), and the maximum error ~~K3 decreased from 0.4833 to 0.27~~ K_3 from 0.4833 to 0.27.

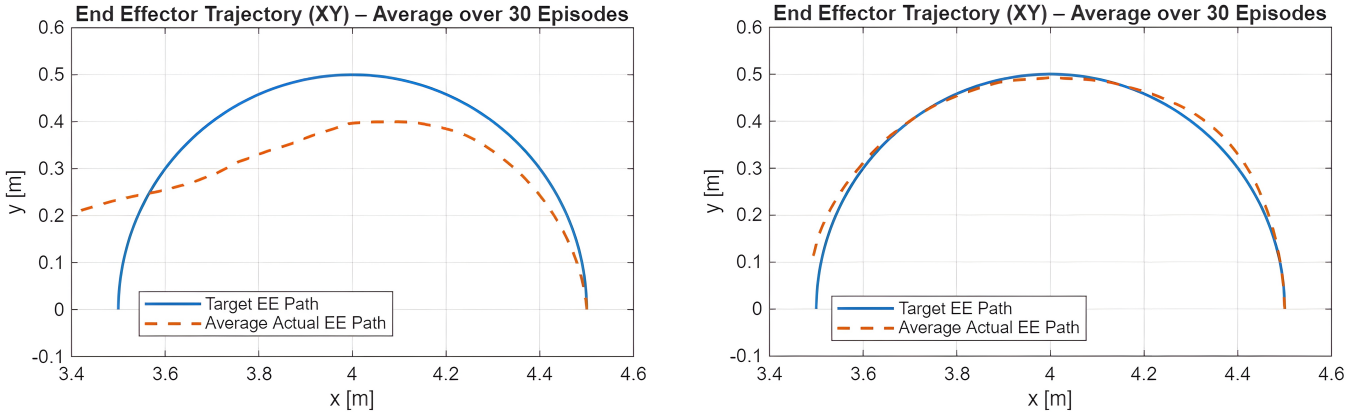
Base stabilization ~~improved markedly~~ improves accordingly (Figure 16): the mean orientation error ~~K4 fell from 0.0667 to 0.022 (factor ≈ 3~~ K_4 decreases from 0.0667 to 0.022 (a factor of approximately three), and the mean angular velocity ~~K8 was halved from 0.095 to 0.050 rad/s~~ K_8 decreases from $0.095\,\text{rad/s}$ to $0.050\,\text{rad/s}$. The optimized agent ~~learned~~ uses arm motions that ~~effectively cancel~~ compensate for momentum transfer to the base.

~~The optimized PPOalso eliminated all safety violations (K5 = K6 = 0 over~~ For the optimized PPO, no safety violations are observed ($K_5 = K_6 = 0$ over the 30 evaluation episodes, compared to ~~5.1% joint-limit violations~~ $K_6 = 5.1\%$ for the default ~~), improved~~ configuration). It also improves torque smoothness (~~K7: 0.6812 →0.351)~~ K_7, $0.6812 \rightarrow 0.351$) and reduces the energy consumption by 16% (K_9, $6.325 \rightarrow 5.314$). These changes are observed simultaneously in the reported evaluation, without an apparent trade-off among the selected KPIs.

The optimized PPO agent satisfies the task objectives considered here in combination in the reported evaluation, namely tracking accuracy, base disturbance, safety constraints, control smoothness, and energy consumption. These results support the use of systematic hyperparameter

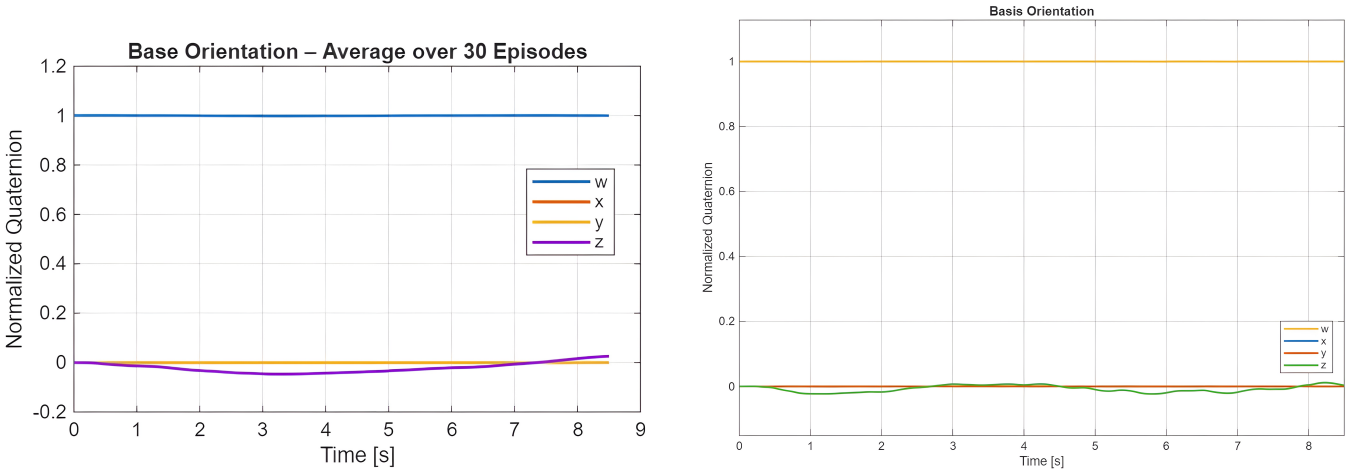
TABLE 7: Performance of PPO agent before and after optimization. The optimized PPO ~~shows substantial improvements in all aspects (~~improves most reported KPIs, with higher rewards, lower errors, ~~zero~~ no observed safety violations in the evaluation episodes, ~~smoother~~ and ~~more efficient~~ smoother control~~)~~.

Metric (KPI)	PPO (Default)	PPO (Optimized)
Mean Episodic Reward (K1)	-2.23	$\mathbf{-0.43}$
Mean Squared EE Tracking Error (K2)	0.0257	**0.0083**
Max EE Tracking Error (K3)	0.4833	**0.27**
Mean Base Orientation Error (K4)	0.0667	**0.022**
Collision Rate (K5)	0	0
Joint Limit Violation Rate (K6)	0.0505 (5.1%)	**0**
Torque Smoothness (K7, jerk)	0.6812	**0.351**
Mean Base Angular Velocity (K8)	0.095	**0.050**
Energy Consumption (K9)	6.325	**5.314**



(a) PPO (Default) (b) PPO (Optimized)

FIGURE 15: Target/actual comparison of the end-effector trajectory in the XY plane.



(a) PPO (Default) (b) PPO (Optimized)

FIGURE 16: ~~Course~~ Time evolution of the base orientation (quaternion components) during a test episode.

tuning and tailored network design for safety-critical space-robot control under the considered conditions.

B. ABLATION STUDY OF REWARD COMPONENTS

To quantify the contribution of each individual term in the composite reward function defined in Equation 1, an ablation study was performed in which the optimized PPO agent was retrained on the circular trajectory with one reward component disabled at a time. Each of the seven weights in the cost term c_t was set to zero in turn while keeping the remaining six at their baseline values ($W_p = 150$, $W_v = 25$, $W_{\text{ori}} = 200$, $W_{wb} = 8$, $W_{vb} = 2$, $W_u = 0.02$, $W_d = 0.06$). In addition, the two positive reward terms were separately disabled by setting $r_{\text{prog}} = 0$ and ~~reduced energy consumptionby 16%~~ ($r_{\text{bonus}} = 0$, respectively, to isolate the contribution of the progress signal and the proximity bonus. All other aspects of the training and evaluation pipeline (hyperparameters, random seeds, episode horizon, and the $N_{eval} = 30$ evaluation protocol) were held identical to the baseline configuration. The resulting KPIs are summarized in Table 8.

TABLE 8: Ablation study of the reward components. Each row corresponds to a separately retrained optimized PPO agent with one reward weight set to zero. KPIs are averaged over $N_{eval} = 30$ evaluation episodes on the circular trajectory.

Variant	K1	K2	K4	K6	K7	K9
Baseline	**−0.43**	**0.0083**	**0.022**	**0**	**0.351**	5.314
$W_p = 0$	−6.48	0.238	0.046	0	0.626	2.868
$W_v = 0$	−0.85	0.022	0.049	0.0015	0.655	3.561
$W_{\text{ori}} = 0$	−23.65	0.017	0.668	0.047	0.588	4.363
$W_{wb} = 0$	−1.17	0.032	0.059	0.0022	0.657	3.554
$W_{vb} = 0$	−1.31	0.039	0.054	0.0004	0.670	3.192
$W_u = 0$	−1.32	0.037	0.060	0	0.618	3.856
$W_d = 0$	−1.10	0.029	0.057	0.0003	0.632	3.724
$r_{\text{prog}} = 0$	−3.65	0.075	0.049	0.0173	0.618	3.365
$r_{\text{bonus}} = 0$	−3.96	0.076	0.057	3.8e−5	0.627	4.078

Dominant weights (W_{ori}, W_p). Removing the base-orientation

penalty has a substantial effect. The mean episodic reward K1 collapses from -0.43 to -23.65, and the mean base-orientation error K4 rises by more than an order of magnitude (from 0.022 to 0.668). Interestingly, the end-effector tracking error K2 even decreases slightly, which reveals that without W_{ori} the agent optimizes EE tracking at the expense of completely losing base stabilization. Because base stability is the primary challenge in free-floating manipulation, this indicates that W_{ori} is central rather than merely fine-tuning the behavior. Removing W_p degrades EE tracking severely (K2 rises by a factor of ≈ 29 and K3 increases approximately $3.5\times$, not shown in the table), driving K1 to -6.48.

Secondary weights (W_v, W_{wb}, W_{vb}). These terms individually have a small but measurable impact: K1 lies in the range $[-1.31, -0.85]$ compared to the baseline -0.43, and tracking accuracy and smoothness are noticeably worse (K2 rises by a factor of 2–5, K7 roughly doubles). They shape the motion quality and the base velocity damping, but are not individually decisive for task completion in this benchmark.

Control-effort weights (W_u, W_d). Disabling the penalties on torque magnitude and torque change modestly increases tracking error (K2) and roughly doubles the jerk metric K7, while K1 remains around -1.3. This confirms that these terms primarily shape actuation smoothness rather than task completion.

Positive reward terms (r_{prog}, r_{bonus}). Removing either of the two positive reward terms causes a pronounced performance drop, with K1 falling to -3.65 and -3.96, respectively, and the tracking error K2 increasing by roughly an order of magnitude (from 0.0083 to ≈ 0.075). Without r_{prog}, the agent loses the dense directional signal that rewards incremental reduction of the position error, and the joint-limit violation rate K6 rises to 1.73%, indicating that the agent tends to execute more aggressive motions when deprived of step-wise progress feedback. Without r_{bonus}, the agent lacks the local attractor near the reference trajectory and does not refine tracking as effectively in the terminal phase. Both terms therefore materially affect training-time convergence and the final tracking accuracy of the policy.

Note on the energy metric K9. K9 ~~: 6.325 →5.314)~~ appears lower in every ablation variant than in the baseline (5.314). This does not indicate improved energy efficiency. Rather, it is an artifact of earlier episode termination when the agent fails (e.g., collision or excessive orientation error), which truncates the integration window. K9 should therefore only be interpreted jointly with K1, K2, and K4. ~~Notably, better tracking~~

Summary. The ablation indicates that W_{ori} and W_p are the two dominant cost components of the composite reward, while the remaining five cost weights contribute refinements in motion quality, base velocity damping, and actuation smoothness. Both positive reward terms (r_{prog} and r_{bonus}) are additionally shown to be important for training-time convergence and final tracking accuracy. Removing any single secondary cost weight still yields a functioning, if degraded, policy, whereas removing W_{ori}, ~~base stability~~W_p, or either of the positive reward terms leads to a markedly degraded controller. This supports the reward design choices reported in section IV and shows that the benchmark conclusions do not rely on fine balancing of low-weight terms.

TABLE 9: One-at-a-time sensitivity analysis of the four dominant reward weights. Each row corresponds to a separately retrained optimized PPO agent with one weight multiplied by $\times 2$ or $\times 0.5$ around its baseline value, all other weights and training settings held constant. KPIs are averaged over $N_{eval} = 30$ evaluation episodes on the circular trajectory.

Variant	K1	K2	K3	K4	K7	K9
Baseline	**−0.43**	**0.0083**	–	**0.022**	**0.351**	5.314
$W_p \times 2$	−4.56	0.0348	0.629	0.073	0.657	3.762
$W_p \times 0.5$	−3.62	0.0494	0.583	0.059	0.611	3.800
$W_v \times 2$	−3.07	0.0246	**0.480**	0.061	0.634	4.212
$W_v \times 0.5$	−3.48	0.0315	0.535	0.064	0.622	3.827
$W_{\text{ori}} \times 2$	−6.89	0.2221	0.907	0.041	0.622	3.059
$W_{\text{ori}} \times 0.5$	−2.82	0.0438	0.786	0.050	0.639	3.546
$W_{wb} \times 2$	−3.41	0.0789	0.746	0.044	0.640	3.283
$W_{wb} \times 0.5$	−3.00	0.0644	0.740	0.045	0.638	3.557

C. SENSITIVITY ANALYSIS OF REWARD WEIGHTS

The ablation study in subsection VII-B identifies which reward components are most influential. A complementary question, raised independently by both reviewers, is whether the conclusions reported in Table 5 are robust against moderate miscalibration of the reward weights: does the PPO agent still converge to a working policy when the dominant cost weights are perturbed, and does the ranking of the algorithms remain qualitatively stable? To answer this, a one-at-a-time sensitivity analysis was performed on the four dominant cost weights identified by the ablation, namely W_p, ~~safety,~~ W_v, W_{ori}, and W_{wb}. Each of these four weights was independently perturbed by a factor of $\times 2$ and ~~energy efficiency were all achieved inparallel, indicating a fundamentally more efficient control strategy.~~ $\times 0.5$ around its baseline value ($W_p = 150$, $W_v = 25$, $W_{\text{ori}} = 200$, $W_{wb} = 8$), while all remaining reward weights, the PPO hyperparameters (as optimized in subsection VII-B), the random seeds, the episode horizon, and the $N_{eval} = 30$ evaluation protocol were held identical to the baseline configuration. This yields eight retrained policies, whose KPIs are summarized in Table 9.

Dominance of W_{ori}. The base-orientation weight exhibits by far the largest sensitivity: K1 varies from -2.82 at $W_{\text{ori}} \times 0.5$ to -6.89 at $W_{\text{ori}} \times 2$, a spread of more than a factor of two. Doubling W_{ori} over-emphasizes base stabilization at the cost of the end-effector tracking objective. K2 rises to 0.222 (a factor of ≈ 27 over the baseline), while K4 remains tight at 0.041. Halving W_{ori}, in contrast, relaxes the base-stabilization pressure without

causing loss of task execution: the agent still tracks the reference ($K2 = 0.0438$) but with twice the baseline base-orientation error ($K4 = 0.050$ vs. baseline 0.022). This is fully consistent with the ablation conclusion that W_{ori} is the dominant cost weight.

Secondary dominance of W_p. The EE position weight is moderately sensitive. K1 ranges from -3.62 to -4.56, and K2 rises by a factor of 4–6 over the baseline in both directions. Both perturbations disturb the balance between EE tracking and base stabilization, but neither causes the policy to break, again consistent with the ablation finding that W_p is the second most influential weight.

Low sensitivity of W_v and W_{wb}. The velocity-error and base-angular-velocity weights show the smallest K1 spread (roughly symmetric around $K1 \approx -3.2$), confirming that these terms shape motion quality and base damping but do not fundamentally alter the task-solvability of the policy. This aligns with the ablation result that completely removing either of these weights still yields a functional, if somewhat degraded, controller.

Safety metrics remain within limits. Across all eight perturbed configurations the collision rate remains at $K5 = 0$ and the joint-limit violation rate stays below $0.53\,\%$ ($K6 \leq 0.0053$). The formal safety-monitoring layer introduced in section V therefore reduces the dependence of constraint enforcement on the exact numerical tuning of the reward weights, which is an important robustness property for an on-orbit-servicing controller. Reward miscalibration degrades task performance but does not compromise the monitored safety limits in the tested cases.

Robustness of the algorithm ranking. Across all eight weight perturbations the optimized PPO agent ~~fulfills all task objectives simultaneously: precise trajectorytracking, minimal base disturbance, zero safety violations~~remains functional, with K1 bounded in $[-6.89, -2.82]$, ~~smooth actuation,~~ $K5 = 0$, and bounded joint-limit violations. Even in the worst-case configuration ($W_{\text{ori}} \times 2$), the resulting EE tracking error ($K2 = 0.222$) remains within the same order of magnitude as the TD3 and SAC baselines reported in Table 5 ($K2 \approx 0.19$ for both, at the nominal weights). Since the other benchmarked agents (TRPO, SAC, TD3, PG, DDPG) are already noticeably weaker at the nominal reward weights, a uniform $\times 2/\times 0.5$ perturbation of any single dominant weight does not plausibly overturn the ranking reported in subsection VII-B. A full cross-evaluation of all six algorithms over the complete eight-configuration grid would multiply the training budget by a factor of six and is beyond the scope of this benchmark. However, the result reported here suggests that the ranking conclusions are not an artifact of a brittle, precisely-tuned operating point.

Note on the energy metric K9. As in the ablation, K9 is consistently lower than the baseline (5.314) across all eight variants. This reflects earlier episode termination in configurations where the agent fails to stabilize (the integration window shortens), not genuine energy efficiency. K9 should therefore be interpreted jointly with K1, K2, and K4.

Summary. The sensitivity analysis supports the reward-design choices reported in section IV: W_{ori} is the most sensitive weight, W_p is secondary, and W_v and ~~reduced energy consumption. This validates the approach of systematic hyperparameter tuning and network tailoring for safety-critical space robot control~~W_{wb} are comparatively insensitive. Task solvability and the monitored safety limits remain satisfied across the full $\times 0.5$ to $\times 2$ range on every tested weight, and the algorithm ranking reported in Table 5 is robust against moderate reward-weight miscalibration.

D. ROBUSTNESS EVALUATION UNDER NON-IDEAL CONDITIONS

To evaluate the robustness of the optimized PPO policy beyond the nominal simulation assumptions, an additional stress-test campaign was performed on the circular trajectory. The already trained optimized PPO policy was evaluated without retraining over $N_{\text{eval}} = 30$ episodes for each condition. Five non-ideal cases were considered, namely sensor noise, reduced actuator authority, an external torque disturbance, parameter uncertainty, and a combined stress case. Sensor noise was injected directly between the observation vector and the RL agent using the MATLAB function $obs_{\text{noisy}} = obs + 0.005\,\text{randn}(\text{size}(obs))$, while the underlying plant dynamics remained unchanged. Actuator saturation was tested by reducing all torque limits to $0.75\,\tau_{\max}$. The external disturbance was implemented as an additive joint torque after the torque-saturation block, with $\tau_{\text{dist},1} = \tau_{\text{dist},2} = 2\,\text{Nm}$ for $2.0 \leq t \leq 2.5\,\text{s}$ and zero disturbance otherwise. Parameter uncertainty was modeled by multiplying all masses, inertias, and joint damping coefficients by a factor of 1.5. The combined case applied all four perturbations simultaneously.

Robustness results. The optimized PPO policy remained within the monitored safety limits in all stress tests. No collisions or joint-limit violations occurred in any condition ($K5 = K6 = 0$). The strongest degradation is observed for the combined stress case and for parameter uncertainty, as expected because both directly change the effective dynamics seen by the fixed policy. Even in the combined case, the controller remains stable, with the mean base-orientation error increasing to $K4 = 0.0447$ while the mean tracking error stays close to the nominal level ($K2 = 0.0078$). Across all non-ideal conditions, K7 increases substantially compared with the nominal optimized PPO result, showing that the policy compensates for perturbations through less smooth torque profiles. The lower K9 values under stress should therefore not be interpreted as genuine efficiency gains. They reflect the changed actuator limits, modified dynamics, and injected disturbances that alter the mechanical-work calculation. Overall, the experiment confirms that the optimized PPO policy degrades gracefully under sensor noise, disturbances, parameter uncertainty, and reduced actuator authority, while the formal safety layer preserves the hard safety constraints.

TABLE 10: Robustness evaluation of the optimized PPO policy under non-ideal simulation conditions. The policy was evaluated without retraining over $N_{\text{eval}} = 30$ episodes per condition on the circular trajectory. K1 is maximized, whereas K2–K9 are minimized.

Condition	K1	K2	K3	K4	K5	K6	K7	K8	K9
Nominal optimized PPO	−0.3709	0.0021	0.1614	0.0227	0	0	0.8401	0.0258	3.9425
Sensor noise	−0.4557	0.0030	0.2164	0.0227	0	0	0.6677	0.0398	3.8498
Reduced saturation ($0.75\,\tau_{\max}$)	−0.4300	0.0028	0.1982	0.0226	0	0	0.6912	0.0373	3.3913
External disturbance	−0.5791	0.0029	0.2052	0.0268	0	0	0.6671	0.0447	3.7735
Parameter uncertainty ($\times 1.5$)	−0.7728	0.0067	0.1775	0.0300	0	0	0.6786	0.0343	2.8879
Combined stress case	−1.3432	0.0078	0.2269	0.0447	0	0	0.6925	0.0402	2.4867

E. PIECEWISE-LINEAR TRAJECTORY

~~While the previous evaluations focused on tracking~~ The previous evaluations consider tracking of a smooth circular end-effector (EE) reference~~, we additionally tested how well~~. To assess how the controller generalizes to ~~linear trajectories. This~~ a different reference geometry, an additional evaluation is performed on a piecewise-linear trajectory. This case is relevant for ~~practical~~ on-orbit manipulation tasks ~~, where~~ in which approach and retreat motions are ~~frequently~~ planned as straight-line segments rather than continuous curves.

Trajectory definition: the linear reference is constructed from the same circle parameters (center and radius) used for the circular benchmark, but the EE is commanded to move along two straight segments connecting three points on that circle: $P_0 = [c_x + r,\ c_y,\ c_z]$, $P_1 = [c_x,\ c_y + r,\ c_z]$, and $P_2 = [c_x - r,\ c_y,\ c_z]$. Using a period T and a switching time $t_1 = T/2$, the piecewise-linear reference position $p_{\text{ref}}(t)$ is

$$p_{\text{ref}}(t) = \begin{cases} P_0 + \frac{t}{t_1}\,(P_1 - P_0)\,, & 0 \le t \le t_1, \\ P_1 + \frac{t-t_1}{T-t_1}\,(P_2 - P_1)\,, & t_1 < t \le T. \end{cases} \tag{5}$$

The reference is sampled at $T_s = 0.01\,\text{s}$, while the PPO policy is updated at $T_{s,\text{agent}} = 0.1\,\text{s}$ (same as in the other experiments).

~~Results:~~ **Results.** Table 11 reports the KPI averages over ~~$N_{eval} = 30$~~ $N_{\text{eval}} = 30$ evaluation episodes for PPO with the default ~~configuration~~ and the optimized ~~configuration, now tested~~ configurations on the piecewise-linear reference. The optimized PPO ~~improves overall task success (K1 increases from -0.4565 to -0.2977~~increases the episodic return (K_1, $-0.4565 \rightarrow -0.2977$), reduces the base attitude disturbance (~~K4 decreases by $\approx 44\%$)~~ (K_4 decreases by approximately 44%, see Figure 18), ~~lowers~~ reduces the residual base angular velocity (~~K8 decreases by $\approx 15\%$~~K_8 decreases by approximately 15%), and ~~significantly reduces~~ reduces the energy consumption (~~K9 decreases by $\approx 35\%$~~K_9 decreases by approximately 35%). The mean and peak ~~end-effector~~ tracking errors (~~K2 and K3) remain almost unchanged~~K_2 and K_3) remain close to those of the default configuration, with a ~~slight improvement in~~ small reduction of the mean error (see Figure 17). ~~However, the~~ The control smoothness metric ~~(K7) becomes worse than~~ K_7, however, is worse than for the default PPO. This ~~suggests~~ indicates that the hyperparameter ~~optimization (highly beneficial~~ configuration tuned for circular tracking ~~)~~ does not fully transfer to piecewise-linear ~~motion profiles. A straightforward~~ references. A direct mitigation is to include a mixture of reference types (circular, linear~~segments~~, and hybrid trajectories) during training ~~(or during hyperparameter search) to explicitly optimize for trajectory generalization~~or during the hyperparameter search.

Both the circular and the piecewise-linear evaluations ~~validate the approach~~ support the use of systematic hyperparameter tuning and ~~network tailoring~~ tailored network design for safety-critical ~~space robot control, while also revealing~~ space-robot control. The piecewise-linear results also show that trajectory-specific ~~optimization may be needed for optimal generalization~~tuning may be required to maintain control smoothness across reference geometries.

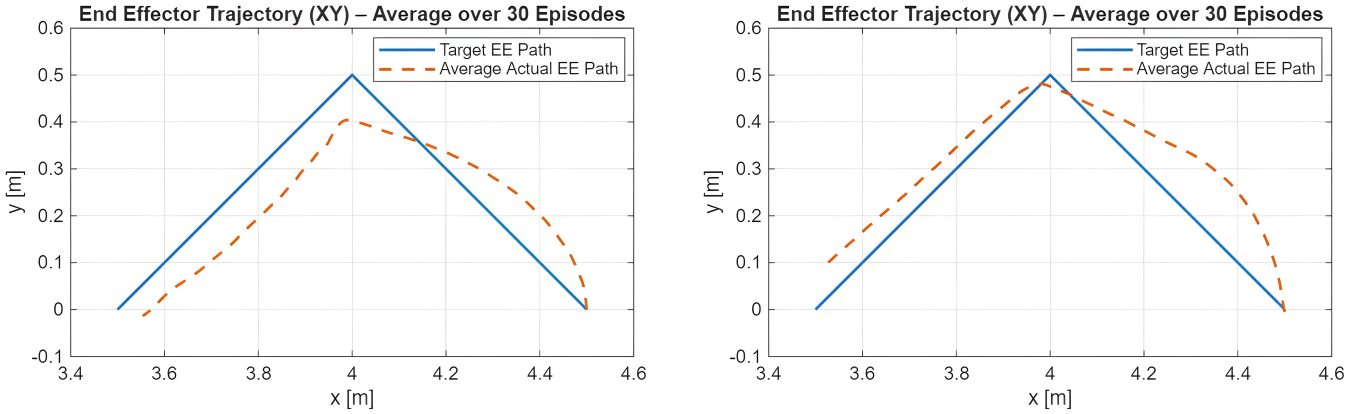
~~In the next section , we will discuss the broader implicationsof these results~~The following section discusses the implications, limitations, and future ~~avenues to bring this work closer to~~ steps toward deployment on physical testbeds for space robotics.

VIII. DISCUSSION AND CONCLUSION

This work has ~~presented a comprehensive study on applying~~ investigated the application of continuous deep reinforcement learning to the control of a free-floating space robotic arm, with an emphasis on safety and performance. The results ~~demonstrate~~ show that an RL-based controller can

TABLE 11: Generalization test on a piecewise-linear end-effector trajectory (two-segment ~~ramp~~path). Values are averaged over $N_{eval} = 30$ episodes. Bold indicates the better value according to the KPI definition (K1 higher is better~~;~~ , K2–K9 lower is better).

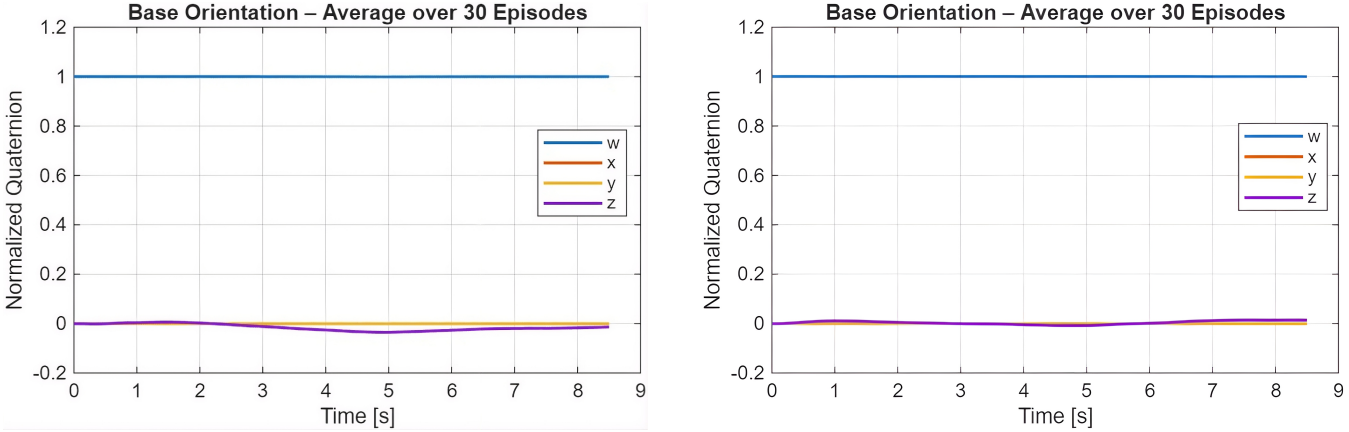
Metric (KPI)	PPO (Default)	PPO (Optimized)
Mean Episodic Reward (K1)	−0.4565	**−0.2977**
Mean Squared EE Position Error (K2)	0.0078	**0.0070**
Max EE Position Error (K3)	0.2037	**0.2036**
Mean Base Orientation Error (K4)	0.0412	**0.0231**
Collision Rate (K5)	0	0
Joint Limit Violation Rate (K6)	0	0
Control Smoothness (K7, jerk)	**0.5520**	0.6839
Mean Base Angular Velocity (K8)	0.0399	**0.0339**
Energy Consumption (K9)	7.7144	**5.0341**



(a) PPO (Default)

(b) PPO (Optimized)

FIGURE 17: Target/actual comparison of the ~~ramp~~ piecewise-linear end-effector trajectory in the XY plane.



(a) PPO (Default)

(b) PPO (Optimized)

FIGURE 18: ~~Course~~ Time evolution of the base orientation (quaternion components) during a test episode with ~~ramp~~ a piecewise-linear trajectory.

~~successfully handle the challenging~~ perform the considered manipulation task in ~~a space environment , achieving high trajectory tracking accuracy while keeping the spacecraft base disturbance to a minimum. The learned agent (especially the optimized PPO policy) was able to coordinate the arm motions in such a way that the~~ the simulated space environment under the stated assumptions. The optimized PPO agent yields a mean squared end-effector ~~closely followed the desired path and the base's unintended motions were small and within acceptable bounds, a key requirement for on-orbit operations. The training process exhibited the typical behavior of RL: an initial exploration phase where performance was poor (and the agent occasionally triggered safety mechanisms), followed by steady improvement as the agent learned from experience, and finally a convergence to a stable policy indicated by decreasing variance in returns. The decreasing variability of the episodic reward and error metrics suggests that the agent's behavior became reliably repeatable and less stochastic as it honed in on an optimal strategy~~ tracking error of $K_2 = 0.0083$, a reduction of 67.7% relative to the default configuration ($K_2 = 0.0257$). The mean base-orientation error decreases by a factor of approximately three (K_4, $0.0667 \rightarrow 0.022$), and the mean base angular velocity is halved (K_8, $0.095\,\mathrm{rad/s} \rightarrow 0.050\,\mathrm{rad/s}$). No safety violations were recorded over the 30 evaluation episodes ($K_5 = K_6 = 0$), compared with a 5.1% joint-limit violation rate for the default configuration. Torque smoothness improves by 48.5% (K_7) and energy consumption decreases by 16% (K_9). These effects are observed together in the reported evaluation rather than as an apparent trade-off between the selected objectives [1], [22].

~~One notable achievement is that the agent can generalize to different initial conditions (within the distribution it was trained on). We tested the final PPO agent on various starting arm configurations and positions along the trajectory. It consistently managed to guide the~~ An additional evaluation was carried out on a piecewise-linear end-effector ~~to the circle and then track it, as long as those initial states were within the training range. This indicates that the policy is~~

not simply memorizing a sequence of actions, but truly learning a feedback control law that is robust to moderate variations in the initial state. However, this generalization only holds within the space of states it experienced. Drastic changes (like a completely different trajectory or a very different initial base orientation) were not tested and would require either retraining or additional training with such scenarios (perhaps via domain randomization [44])reference to assess trajectory generalization. The optimized PPO agent outperforms the default configuration on most metrics, increasing the return K_1 from -0.4565 to -0.2977, reducing the base-orientation error K_4 by 44%, and reducing the energy K_9 by 35%, which indicates partial transfer of the learned policy. The control-smoothness metric K_7, however, is worse than for the default configuration, showing that the hyperparameter configuration tuned for circular tracking does not fully transfer to piecewise-linear references. This motivates either trajectory-diverse training or trajectory-specific tuning to support generalization across reference geometries [44].

Practical Implications:. The success of PPO (and TRPO) in this domain suggests that policy gradient methods are well-suited for complex, performance of PPO and TRPO on this task indicates that policy-gradient methods with constrained updates are suitable candidates for continuous control tasks with coupled dynamics strong dynamic coupling and safety constraints. In particular, PPO's ability to handle continuous action spaces and its stable update rule (The clipped surrogate objective) made it robust in the face of the highly nonlinear dynamics of the of PPO provides bounded policy updates under the nonlinear free-floating robotdynamics. The integrated safety components (collision avoidance, etc.) effectively acted as a form of "shielding" during training, they prevented the agent from exploring disastrously unsafe actions by cutting those episodes short [20], [24], [25]. This, combined monitor, joint-limit assertion, torque-bound verification) act as a runtime shield during training. They terminate episodes in which the policy would violate a monitored constraint and apply a terminal penalty [20], [24], [25]. Combined with the penalty-driven learning, means the final policy inherently respects those constraints (since it learned quickly that violating them yields no reward). For real missions, this is very promising: it implies we can embed safety checks reward, this leads to a final policy that respects the monitored constraints in the reported evaluation. For deployment on a physical system, this suggests that safety checks embedded in the training simulator and reasonably expect the trained policy to abide by them when deployed (assuming the real system is may remain useful at runtime, provided that the deployed system stays within the simulator's fidelity) 's fidelity range [24], [31].

Limitations and Future Work:. Several limitations should be acknowledged. First, a sim-to-real gap remains: unmodeled effects such as flexible dynamics, . Beyond the idealizing assumptions stated in section III (no sensor noise, actuator latency, and external disturbances(e.g., gravity gradients, magnetic torques)could degrade real-world performance no latency, rigid body model, no external disturbances), real deployment introduces additional challenges. These are (a) domain shift due to inaccuracies in the identified mass and inertia parameters, joint friction, and link flexibility not captured in the URDF model, (b) onboard compute constraints, as the learned neural network policy must execute in real time on embedded space-qualified hardware (typically ARM-based processors with limited floating-point throughput), whereas training was performed on a workstation-class CPU, and (c) sensor-actuator latency, where non-negligible delays between state measurement and torque application can destabilize a policy trained under the zero-latency assumption. Mitigating these effects requires domain randomization over simulation parameters, latency injection during training, and ultimately hardware-in-the-loop (HIL) validation [24], [44]. Second, the RL agent exhibits hyperparameter sensitivity—the . The tuned PPO configuration is task-specific and may require re-tuning for different trajectories or mission scenarios. Third, the state representation was manually designed; learned . Learned representations or recurrent architectures (LSTM) might improve performance under partial observability. Fourth, the safety monitoring covers targeted critical properties but does not constitute exhaustive formal verification of the full mission [23], [24]. Finally, no hardware testing was performed; real-time . Real-time computation limits, communication delays, and physical interactions remain to be validated on physical testbeds.

Future work should address these gaps through: in several directions. These are (i) domain randomization over simulation parameters to improve robustness for and latency injection to improve sim-to-real transfer [44] ; (robustness [44] (see fig. 19), (ii) extension to more complex tasks such as dual-arm coordination, on-orbit assembly, or active debris removal; , (iii) advanced RL architectures (recurrent policies, attention mechanisms) and safe RL algorithms with explicit constraint handling via Lagrange multipliers or barrier functions [24], [25], [31], [32]; , and (iv) hardware-in-the-loop validation e.g. on our on the authors' testbed for In-Space Robotized Services (IROS) with virtually simulated microgravity and predicted energy consumption (see **??**fig. 19).

Conclusion:. In conclusion, this research demonstrates a viable framework for applying This work has presented a framework that combines deep reinforcement learningto complex space robotic systems. We showed that by combining high-fidelity modeling, multiple RL algorithm evaluations, and integrated formal safety checks, one can develop a control policy that achieves high precision and reliability in a challenging , multibody simulation, and formal-methods-based safety monitoring for Cartesian path tracking of a free-floating manipulation task. The comparative analysis revealed PPO as an effective solution, and further optimizations space robot. A comparison of six

continuous-action RL algorithms under unified conditions identified PPO as the algorithm with the most favorable trade-off between tracking accuracy, base disturbance, and training stability for the considered task. Subsequent hyperparameter tuning and network tailoring of the PPO agent yielded a policy that~~meets stringent trajectory and safety requirements. Importantly, the use of formal methods alongside RL is a novel contribution, it addresses the~~, relative to the default PPO configuration, reduces the tracking error by 67.7%, reduces the base-orientation error by a factor of approximately three, shows no joint-limit or collision violations over 30 evaluation episodes, and reduces the energy consumption by 16%. The integration of monitored safety constraints and torque-bound verification into the training loop helped keep these results within the defined safety envelope, addressing part of the typical black-box nature of learned controllers~~by adding verifiable guarantees for certain behaviors. This synergy of model-based and data-driven methods is very promising for space systems, where safety cannot be compromised~~. These findings, grounded in quantitative experimental evidence, support further investigation of safety-augmented deep RL for safety-critical space manipulation tasks.

The ~~work also provides insights into~~ results also illustrate the influence of algorithm choice and parameter tuning on learning performance for ~~physical systems. Our findings reinforce the notion that there is no one-size-fits-all in RL~~ , careful selection and tuning of the agent (and network structure) can significantly impact the outcome, ~~even more so than we might expect from generic benchmarks. For practitioners, this means that investing time in benchmarking algorithms (as we did) can pay off in identifying the right approach for the problem at hand.~~ the considered task. They support the view that no single RL algorithm is uniformly best across applications: under unified conditions, the choice of algorithm, network architecture, and hyperparameter setting visibly affects the achieved tracking accuracy and base stability. Systematic benchmarking of candidate algorithms is therefore a practical step in the selection of an appropriate RL controller for a given task.

~~Finally, while challenges~~ While several open problems remain for real-world deployment, ~~this study lays a foundation. It suggests that future space robots, tasked with servicingsatellites, assembling structures, or handling space debris~~ the proposed framework provides a basis for further investigation. Possible application areas include satellite servicing, ~~could employ~~ on-orbit assembly, and active debris removal, where learning-based controllers ~~that adapt to complex dynamics and uncertainties, all while operating safely within predefined limits. We envision an architecture where an RL policy drives the robot's motions, supervised by a layer of formal logic~~ can adapt to uncertain dynamics within predefined operational limits. The combination of an RL policy with a separate runtime layer that monitors and~~vetoes any unacceptable actions (though, ideally, the policy has learned not to propose such actions in the first place). Such an approach could provide the flexibility and efficiency of learned behaviors with the trust and guarantee of model-based safety, a combination well-suited to the unforgiving environment of space~~ , where required, overrides commands violating the specified constraints is one realization of such an architecture. In this combination, the learned policy provides adaptability, while the monitoring layer enforces the specified safety-critical properties [1], [25].

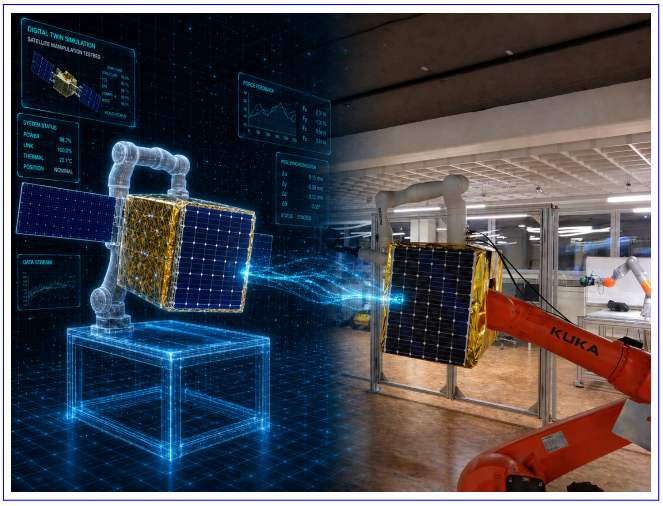


FIGURE 19: ~~Our testbed~~ Gradual hybrid testbeds integrate AI-driven digital twins (left) with physical robotic systems (right), providing a framework for ~~In-Space Robotized Servicesstudying~~ policy optimization and domain-randomization strategies for sim-to-real transfer. Illustration created using a picture of the physical setup at Frankfurt University of Applied Sciences and with the help of GPT-Image-2.

~~Ultimately, our work demonstrates a step in that direction, confirming that with the right algorithms and safeguards, "autonomous learning-based control" can be a powerful paradigm for future space robotic missions. The results encourage continued~~ The present study contributes one step in this direction. Future research at the intersection of robotics, reinforcement learning, and formal methods ~~to further unlock the potential of intelligent space systems~~is required to extend the present results to physical deployment and to broader mission scenarios.

ACKNOWLEDGMENT

The authors used DeepL Translator/Write [45], [46] and OpenAI ChatGPT [47] solely for language editing and grammar enhancement. All technical content and conclusions were created and verified by the authors.

REFERENCES

[1] A. Flores-Abad, O. Ma, K. Pham, and S. Ulrich, "A review of space robotics technologies for on-orbit servicing," Prog. Aerosp. Sci., vol. 68, pp. 1–26, 2014, doi: 10.1016/j.paerosci.2014.03.002.

[2] K. Nanos and E. G. Papadopoulos, "On the dynamics and control of free-floating space manipulator systems in the presence of angular momentum," Front. Robot. AI, vol. 4, p. 26, 2017, doi: 10.3389/frobt.2017.00026.

[3] E. Papadopoulos, I. Kaliakatsos, and D. Psarros, "Minimum fuel techniques for a space robot simulator with a reaction wheel and PWM thrusters," in Proc. Eur. Control Conf. (ECC), 2007, pp. 3391–3398.

[4] E. Papadopoulos and A. Abu-Abed, "Design and motion planning for a zero-reaction manipulator," in Proc. IEEE Int. Conf. Robot. Autom. (ICRA), 1994, pp. 1554–1559, doi: 10.1109/ROBOT.1994.351367.

[5] M. Mühlbauer, M. Chalon, M. Ulmer, and A. Albu-Schäffer, "Software for the SpaceDREAM robotic arm," in Proc. IEEE Aerosp. Conf., 2025.

[6] T. Rybus, "Robotic manipulators for in-orbit servicing and active debris removal: Review and comparison," Prog. Aerosp. Sci., vol. 151, p. 104550, 2024.

[7] E. Guiffo Kaigom, T. Jung, and J. Roßmann, "Optimal motion planning of a space robot with base disturbance minimization," in Proc. 11th Symp. Adv. Space Technol. Robot. Autom., 2011.

[8] E. Guiffo Kaigom, "Deep generative and discriminative digital twin endowed with variational autoencoder for unsupervised predictive thermal condition monitoring of physical robots in Industry 6.0 and Society 6.0," IFAC-PapersOnLine, vol. 59, pp. 7–12, 2025.

[9] Z. Huang, G. Chen, Y. Shen, R. Wang, C. Liu, and L. Zhang, "An obstacle-avoidance motion planning method for redundant space robot via reinforcement learning," Actuators, vol. 12, no. 2, p. 69, 2023.

[10] A. Al Ali and Z. Zhu, "Reinforcement learning for path planning of free-floating space robotic manipulator with collision avoidance and observation noise," Front. Control Eng., vol. 5, p. 1394668, 2024.

[11] M. D'Ambrosio, L. Capra, A. Brandonisio, and M. Lavagna, "Failure management via deep reinforcement learning for robotic in-orbit servicing," in Proc. SPAICE—1st Joint ESA/IAA Conf. AI Space, 2024, pp. 335–339.

[12] A. Al Ali, J. Shi, and Z. Zhu, "Path planning of 6-DOF free-floating space robotic manipulators using reinforcement learning," Acta Astronaut., vol. 224, pp. 367–378, 2024.

[13] Y. Hu, D. Zhou, W. Yao, X. Shao, and G. Sun, "Deep reinforcement learning-based trajectory planning with continuous pose representation for 6-DoF free-floating space robot," Aerosp. Sci. Technol., p. 110540, 2025.

[14] O. Zhang, Z. Liu, X. Shao, W. Yao, L. Wu, and J. Liu, "Learning-based task space trajectory planning framework with pre-planning and post-processing for uncertain free-floating space robots," IEEE Trans. Aerosp. Electron. Syst., early access, 2025.

[15] Y. Wei, X. Bai, and H. Lu, "Trajectory planning of free-floating space robot for non-cooperative tumbling target capture based on deep reinforcement learning," Robotica, vol. 43, pp. 2674–2692, 2025.

[16] Y. Wu, Z. Yu, C. Li, M. He, B. Hua, and Z. Chen, "Reinforcement learning in dual-arm trajectory planning for a free-floating space robot," Aerosp. Sci. Technol., vol. 98, p. 105657, 2020.

[17] J. Schulman, S. Levine, P. Abbeel, M. Jordan, and P. Moritz, "Trust region policy optimization," in Proc. Int. Conf. Mach. Learn. (ICML), 2015, pp. 1889–1897.

[18] S. Fujimoto, H. van Hoof, and D. Meger, "Addressing function approximation error in actor-critic methods," in Proc. Int. Conf. Mach. Learn. (ICML), 2018, pp. 1587–1596.

[19] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, "Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor," in Proc. Int. Conf. Mach. Learn. (ICML), 2018, pp. 1861–1870.

[20] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," arXiv:1707.06347, 2017.

[21] R. J. Williams, "Simple statistical gradient-following algorithms for connectionist reinforcement learning," Mach. Learn., vol. 8, pp. 229–256, 1992.

[22] R. S. Sutton and A. G. Barto, Reinforcement Learning: An Introduction, 2nd ed. Cambridge, MA, USA: MIT Press, 2018.

[23] MathWorks, "Simulink Design Verifier," ~~The MathWorks, Inc., Natick, MA, USA,~~ MATLAB & Simulink Documentation, 2024. [Online]. Available: https://www.mathworks.com/help/sldv/

[24] J. García and F. Fernández, "A comprehensive survey on safe reinforcement learning," J. Mach. Learn. Res., vol. 16, pp. 1437–1480, 2015.

[25] M. Alshiekh, R. Bloem, R. Ehlers, B. Könighofer, S. Niekum, and U. Topcu, "Safe reinforcement learning via shielding," in Proc. AAAI Conf. Artif. Intell., 2018, pp. 2669–2678.

[26] Open Robotics, "URDF (Unified Robot Description Format)," ROS 2 Documentation, 2024. [Online]. Available: https://docs.ros.org/en/rolling/Tutorials/Intermediate/URDF/URDF-Main.html

[27] MathWorks, "Import URDF models," MATLAB & Simulink Documentation, 2024. [Online]. Available: https://www.mathworks.com/help/sm/ug/urdf-import.html

[28] MathWorks, "smimport—Import a CAD, URDF, or Robotics System Toolbox model," MATLAB & Simulink Documentation, 2024. [Online]. Available: https://www.mathworks.com/help/sm/ref/smimport.html

[29] MathWorks, "Mechanism Configuration," MATLAB & Simulink Documentation, 2024. [Online]. Available: https://www.mathworks.com/help/sm/ref/mechanismconfiguration.html

[30] J. Kober, J. A. Bagnell, and J. Peters, "Reinforcement learning in robotics: A survey," Int. J. Robot. Res., vol. 32, no. 11, pp. 1238–1274, 2013.

[31] J. Achiam, D. Held, A. Tamar, and P. Abbeel, "Constrained policy optimization," in Proc. Int. Conf. Mach. Learn. (ICML), 2017, pp. 22–31.

[32] A. D. Ames, S. Coogan, M. Egerstedt, G. Notomista, K. Sreenath, and P. Tabuada, "Control barrier functions: Theory and applications," in Proc. Eur. Control Conf. (ECC), 2019, pp. 3420–3431.

[33] F. Leutert et al., "AI-enabled cyber–physical in-orbit factory—AI approaches based on digital twin technology for robotic small satellite production," Acta Astronaut., vol. 217, pp. 1–17, 2024.

[34] M. Mühlbauer et al., "Towards an in-orbit cubesat factory," in Proc. 76th Int. Astronaut. Congr. (IAC), 2025.

[35] O. Maqbool and J. Roßmann, "Systems aware scenario engineering for life cycle-spanning verification and validation of complex CPS," Authorea Preprints, 2025. [Online]. Available: https://www.authorea.com

[36] J. Virgili-Llop, "SPART: Spacecraft Robotics Toolkit," 2017. [Online]. Available: https://github.com/NPS-SRL/SPART

[37] E. Ribeiro, R. Mendes, M. Terra, and V. Grassi, "Dynamic parameter identification of a 7-DoF lightweight robot manipulator using probabilistic differential optimization," in Proc. IEEE Int. Conf. Syst., Man, Cybern. (SMC), 2022, pp. 1917–1923.

[38] MathWorks, "6-DOF Joint (Simscape Multibody)," MATLAB & Simulink Documentation, 2024. [Online]. Available: https://www.mathworks.com/help/sm/ref/6dofjoint.html

[39] MathWorks, "Spatial Contact Force," MATLAB & Simulink Documentation, 2024. [Online]. Available: https://www.mathworks.com/help/sm/ref/spatialcontactforce.html

[40] MathWorks, "Solver Configuration," MATLAB & Simulink Documentation, 2024. [Online]. Available: https://www.mathworks.com/help/simscape/ref/solverconfiguration.html

[41] MathWorks, "Fixed-step size (fundamental sample time)," MATLAB & Simulink Documentation, 2024. [Online]. Available: https://www.mathworks.com/help/simulink/gui/fixedstepsizefundamentalsampletime.html

[42] MathWorks, "Reinforcement Learning Toolbox—rlPPOAgent," MATLAB & Simulink Documentation, 2024. [Online]. Available: https://www.mathworks.com/help/reinforcement-learning/ref/rl.agent.rlppoagent.html

[43] T. P. Lillicrap et al., "Continuous control with deep reinforcement learning," in Proc. Int. Conf. Learn. Represent. (ICLR), 2016.

[44] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaremba, and P. Abbeel, "Domain randomization for transferring deep neural networks from simulation to the real world," in Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS), 2017, pp. 23–30.

[45] DeepL SE, "DeepL Translator," [Online]. Available: https://www.deepl.com/de/translator

[46] DeepL SE, "DeepL Write," [Online]. Available: https://www.deepl.com/de/write

[47] OpenAI, "ChatGPT," [Online]. Available: https://chat.openai.com

[48] E. Todorov, T. Erez, and Y. Tassa, "MuJoCo: A physics engine for model-based control," in Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS), 2012, pp. 5026–5033.

[49] N. Koenig and A. Howard, "Design and use paradigms for Gazebo, an open-source multi-robot simulator," in Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS), vol. 3, 2004, pp. 2149–2154.

[50] M. Körber, J. Lange, S. Rediske, S. Steinmann, and R. Glück, "Comparing popular simulation environments in the scope of robotics and reinforcement learning," arXiv:2103.04616, 2021.

[51] J. Collins, S. Chand, A. Vanderkop, and D. Howard, "A review of physics simulators for robotic applications," IEEE Access, vol. 9, pp. 51416–51431, 2021.

PATRICK S. ERMISCH Patrick S. Ermisch was born in Frankfurt am Main, Germany, on December 18, 2002. He received the Abitur degree and is currently pursuing the B.Eng. degree in mechatronics at Frankfurt University of Applied Sciences, Frankfurt, Germany, with an expected graduation in 2026. His major field of study is mechatronics.

He has gained practical experience through an internship at SEW-EURODRIVE, where he contributed to the further development of a mobile application for the manual control of logistics robots. In addition, he worked as a tutor for Physics I and Physics II and as a student research assistant, supporting research on particle simulation and the development of an application for visualization and simulation purposes. His research interests include reinforcement learning for robotics, space robotic systems, and simulation-based control methods.

Mr. Ermisch has been listed on the Dean's List of Frankfurt University of Applied Sciences multiple times in recognition of his academic performance.

ERIC GUIFFO KAIGOM was born in Douala, Cameroon. He received the Dipl.-Ing. and Dr.-Ing. (summa cum laude) degrees in electrical engineering from RWTH Aachen University, ~~Aachen, Germany, in 2010 and 2016, respectively~~Germany. He was a Research Associate with the Institute for Man-Machine Interaction at RWTH Aachen University, where he led the ~~"Intelligent Robots"~~ "Intelligent Robots" team and coordinated technology development and transfer activities with partners in ~~robotized~~ robotics-based manufacturing and on-orbit servicing, including projects ~~within~~ funded under the European Union's Horizon 2020 ~~EU~~ and In-Space Automation programs. He then worked in ~~the industry as a~~ industry as Head of Emerging Technologies~~, serving e.g. as Co-PI for the design and acquisition of a data- and AI-driven flagship project. He~~. Dr. Kaigom is currently a Full Professor with ~~the~~ Frankfurt University of Applied Sciences, ~~Frankfurt,~~ Germany. His research interests include ~~Robotics, Digital Twin, and AI/ML. Dr. Kaigom serves the academic community, currently e.g. as an Associate Editor for IEEE Transactions on Technology and Society and reviewer for several journals (IEEE Trans. Ind. Inform., IEEE Trans. Cybern., Journal of Manufacturing Systems, Engineering Applications of Artificial Intelligence, IEEE RA-L, etc.). He is a recipient of best paper and academic awards, including the Borchers' Medal of the RWTH Aachen University~~robotics, digital twins, and artificial intelligence / machine learning.